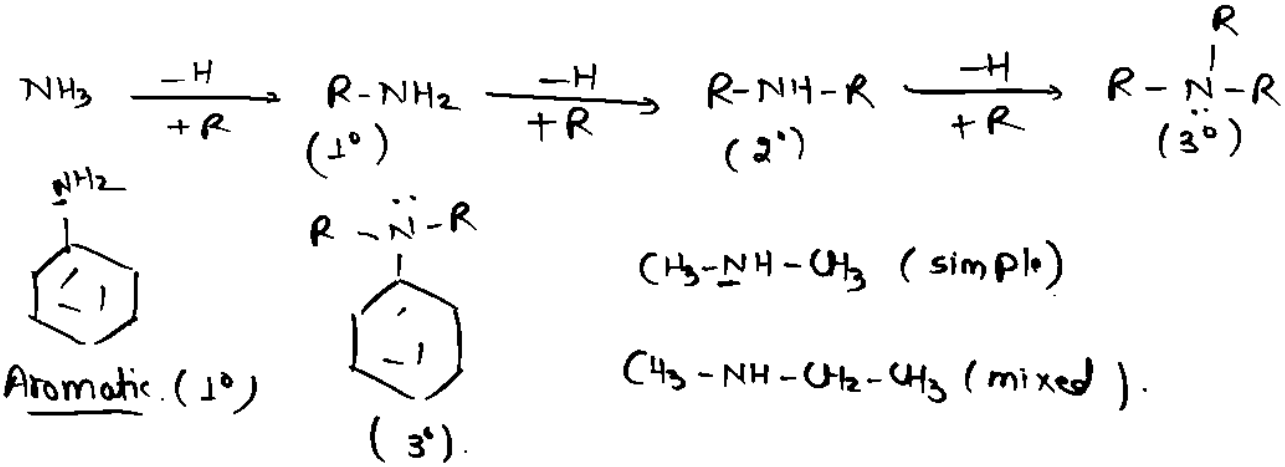
AMINES

CHAPTER SIGNIFICANCE FOR JEE

The Amines chapter is considered a high-output, low-effort chapter for JEE Main and Advanced. While chapters like Aldehyde, Ketone, and Carboxylic Acid are large and complex, Amines is relatively small but provides a significant number of marks in the examination. It is a vital part of the organic chemistry syllabus and serves as the primary section for studying nitrogen-containing compounds.

DEFINITION OF AMINES

Amines are viewed as derivatives of Ammonia (NH_3). They are formed by the replacement of one, two, or all three hydrogen atoms of the ammonia molecule by alkyl (R) or aryl (Ar) groups.



CLASSIFICATION BY DEGREE

Amines are classified as primary, secondary, or tertiary based on the number of hydrogen atoms replaced in the NH_3 molecule. This classification is unique compared to alcohols or alkyl halides, as it depends on the substitution of the Nitrogen atom itself.

Degree	1° (Primary)	2° (Secondary)	3° (Tertiary)
Structure	$R-NH_2$	R-NH-R	R_3N

Common Misconception: In Alcohols, a 2° alcohol is an -OH on a 2° carbon. In Amines, a 2° amine is NOT necessarily on a 2° carbon; it MUST have two carbon groups attached directly to the Nitrogen atom.

QUATERNARY AMMONIUM SALTS

Nitrogen has a lone pair of electrons even in 3° amines (R_3N:). If this lone pair is used to form a coordinate bond with another alkyl group (R), it forms a Quaternary Ammonium Salt.

General Formula: $[R_4N]^+ R^-$

In these salts, the nitrogen atom carries a formal positive charge and is bonded to four carbon groups.

ALIPHATIC VS. AROMATIC AMINES

Classification based on the nature of the groups attached to Nitrogen:

ALIPHATIC AMINES
The Nitrogen atom is attached to alkyl groups (open chains or non-aromatic rings).
Example: CH_3NH_2 (Methylamine).

AROMATIC AMINES
The Nitrogen atom is directly attached to one or more aromatic (aryl) rings.
Example: Aniline ($C_6H_5NH_2$), where the $-NH_2$ group is directly bonded to the benzene ring.
Classification within Aromatic:
- 1° Aromatic: Aniline (1 ring on N).
- 3° Aromatic: Triphenylamine (3 rings on N).

SIMPLE VS. MIXED AMINES
Secondary and tertiary amines can be further categorized based on the symmetry of the groups attached to the Nitrogen.

Type	Definition	Example
Simple (Symmetrical)	All alkyl or aryl groups on Nitrogen are the same.	$CH_3)_2NH$ (Dimethylamine)
Mixed (Unsymmetrical)	The alkyl or aryl groups on Nitrogen are different.	CH_3-NH-C_2H_5 (Ethylmethylamine)

FORMULA SHEET
GENERAL STRUCTURES
- Ammonia: NH_3
- Primary Amine: RNH_2
- Secondary Amine: R_2NH
- Tertiary Amine: R_3N
- Quaternary Salt: $R_4N^+ X^-$

COMMON TRAPS

Mistake	Why Students Make It	Correct Thinking
Degree confusion	Thinking of carbon degrees	Count carbons on N, not the degree of the R-carbon.
Amine vs Amide	Seeing a C=O nearby	If Nitrogen is next to C=O, it is an Amide, not an Amine.

ELECTRONIC STRUCTURE AND HYBRIDIZATION
AMINE DERIVATION AND HYBRIDIZATION
Amines are nitrogen-containing organic compounds derived from ammonia (NH_3). In any amine (whether primary, secondary, or tertiary), the nitrogen atom is sp^3 hybridized. The electronic configuration of the nitrogen atom in this state involves four sp^3 hybrid orbitals.

SUMMARY OF ELECTRONIC STATE
- Central Atom: Nitrogen (N).
- Steric Number: 3 Bond Pairs + 1 Lone Pair = 4.
- Hybridization: sp^3.

GEOMETRY AND MOLECULAR SHAPE
IDEAL VS. ACTUAL GEOMETRY
According to the hybridization theory, four sp^3 orbitals should ideally lead to a tetrahedral geometry. However, the molecular shape is defined only by the positions of the atoms (bond pairs), not the lone pair.

TRIGONAL PYRAMIDAL SHAPE
Because one of the four positions is occupied by a lone pair, the resulting molecular shape of an amine is Trigonal Pyramidal. The nitrogen atom sits at the apex of the pyramid, and the three attached groups (R or H) form the triangular base.

BOND ANGLE DEVIATION FROM IDEAL ANGLE

The ideal bond angle for a perfectly tetrahedral molecule (like CH_4) is 109.5°. In amines, the presence of the lone pair causes a deviation from this ideal angle. The bond angle in amines is always less than (<) 109.5°.

JEE ADVANCED INSIGHTS

LONE PAIR REACTIVITY

The lone pair is not just responsible for the shape; it is the reason amines behave as nucleophiles and bases. This lone pair is available for donation to electrophiles (like H^+), leading to the formation of quaternary ammonium salts where the geometry finally becomes perfectly tetrahedral (as there are then 4 bond pairs and 0 lone pairs).

STERIC IMPACT ON ANGLES

While the lone pair reduces the angle, very bulky alkyl groups (like tertiary butyl groups) can push back against this "squeezing," sometimes causing the bond angle to increase slightly compared to simpler amines, though it typically remains near or below the ideal tetrahedral value.

HIGH-YIELD REVISION SHEET

MUST REMEMBER:

- Hybridization: sp^3.
- Steric Number: 4 (3 Bond Pairs + 1 Lone Pair).
- Geometry: Tetrahedral.
- Shape: Trigonal Pyramidal.
- Bond Angle: Less than 109.5° (typically approx. 108°).

LAST-MINUTE REVISION POINTS:

- The lone pair occupies more space than a bond pair, leading to compressed bond angles.
- Nitrogen is at the apex of the pyramid.
- In quaternary ammonium salts (R_4N^+), nitrogen is still sp^3 but the shape is Tetrahedral and the angle is approx. 109.5°.

METHOD OF PREPARATION OF AMINES

1. REDUCTION OF NITRO COMPOUNDS

Nitro compounds ($R\text{-}NO_2$) are reduced to primary amines ($R\text{-}NH_2$) by replacing oxygen atoms with hydrogen atoms. This is a straightforward and highly effective method for both aliphatic and aromatic amines.

REAGENTS AND CONDITIONS

* Catalytic Hydrogenation: H_2 in the presence of finely divided Palladium (Pd), Platinum (Pt), or Nickel (Ni).
* Chemical Reduction in Acidic Medium: Metal and concentrated Hydrochloric acid (HCl).
  * Tin and HCl (Sn + HCl)
  * Iron and HCl (Fe + HCl)

ADVANTAGE OF FE + HCL

This is the preferred industrial method. During the reaction, $FeCl_2$ is formed as a byproduct. $FeCl_2$ undergoes hydrolysis to release HCl back into the reaction mixture.

$$Fe + 2HCl \rightarrow FeCl_2 + 2[H]$$
$$FeCl_2 + 2H_2O \rightarrow Fe(OH)_2 + 2HCl$$

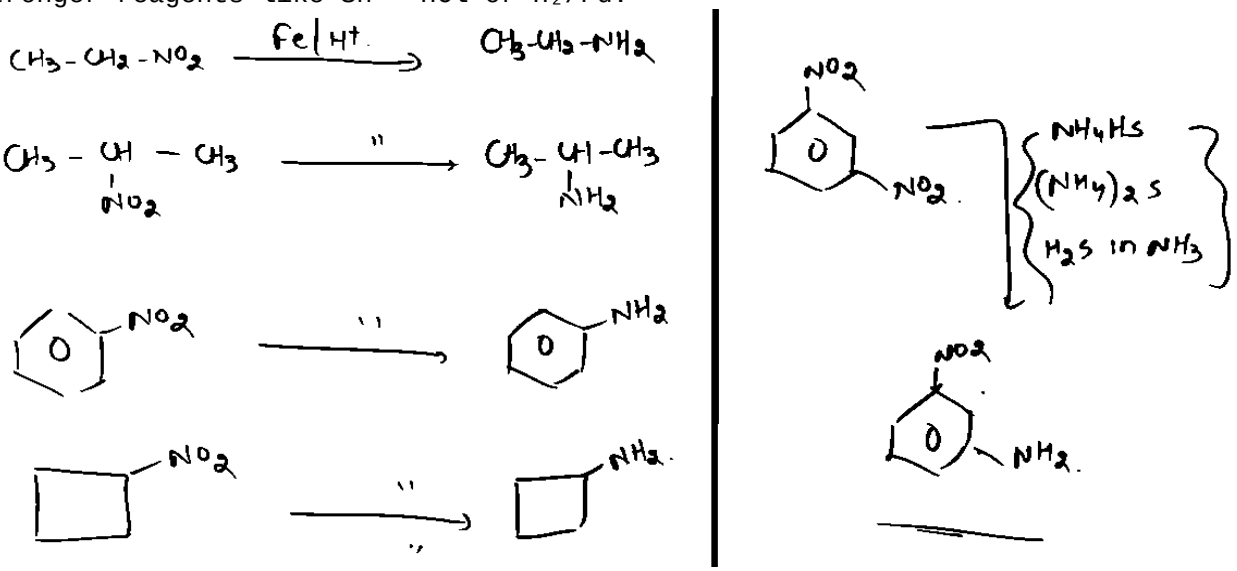
Because HCl is regenerated, only a small amount of initiatory acid is required, making the process economical.

SELECTIVE REDUCTION

When a molecule contains two nitro groups (dinitro compounds), selective reduction of only one nitro group can be achieved using specific reagents:

* Ammonium hydrogen sulphide: $(NH_4)HS$
* Ammonium sulphide: $(NH_4)_2S$
* H_2S in NH_3

This prevents both groups from reducing simultaneously, which would happen with stronger reagents like Sn + HCl or H_2/Pd.



2. AMMONOLYSIS OF ALKYL HALIDES (HOFFMANN'S METHOD)

The reaction involves the cleavage of the Carbon-Halogen (C-X) bond by an ammonia molecule. It is a Nucleophilic Substitution reaction where Ammonia (NH_3) acts as the nucleophile.

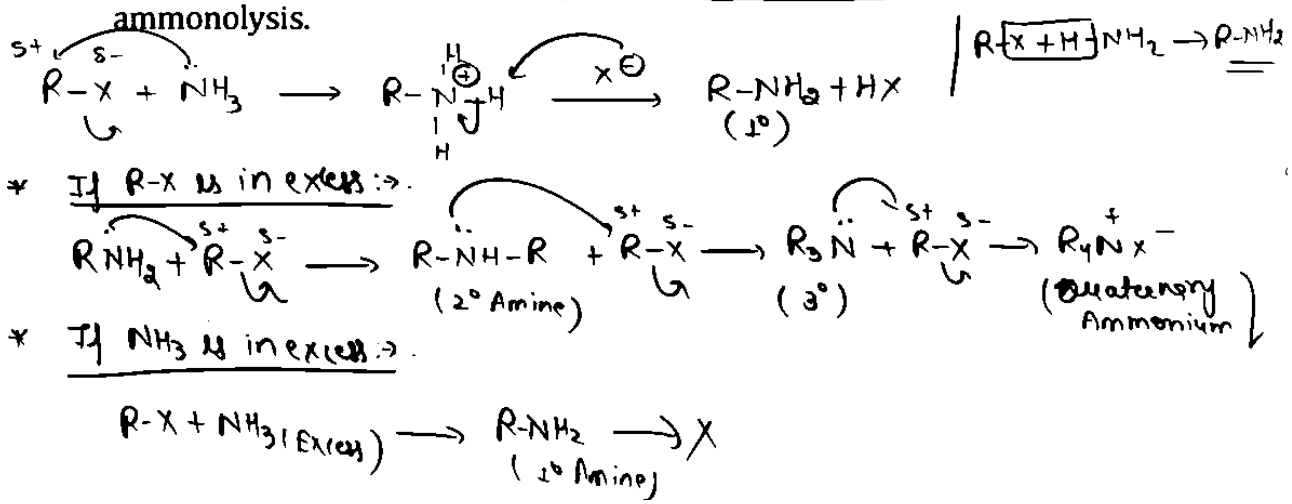
Mechanism steps:

1. NH_3 attacks the electrophilic carbon of R-X, displacing the halide (X^-).
2. A substituted ammonium salt is formed: $[R\text{-}NH_3^+ X^-]$.
3. X^- attacks the hydrogen of amine, and $R\text{-}NH_2$ is formed with HX.

REACTION WITH EXCESS REAGENTS

* Excess Alkyl Halide: The reaction does not stop at the primary amine. Since the nitrogen in the resulting amine still has a lone pair, it continues to attack the excess R-X. This leads to a mixture of:
  1° Amine → 2° Amine → 3° Amine → 4° Quaternary Ammonium Salt $[R_4N^+ X^-]$.
* Excess Ammonia: If NH_3 is used in large excess, it ensures that every R-X molecule is attacked by NH_3 rather than the newly formed amine. This makes the Primary Amine (1°) the major product.

➢ The process of cleavage of the C-X bond by ammonia molecule is known as ammonolysis.



Imp points.
- Ammonolysis has the disadvantage of yielding a mixture of primary, secondary, tertiary and quaternary amines.
- However, primary amine is obtained as a major product by taking large excess of ammonia.
- The order of reactivity of halides with amines is R-I > R-Br > R-Cl and it is the order of rate of reaction.

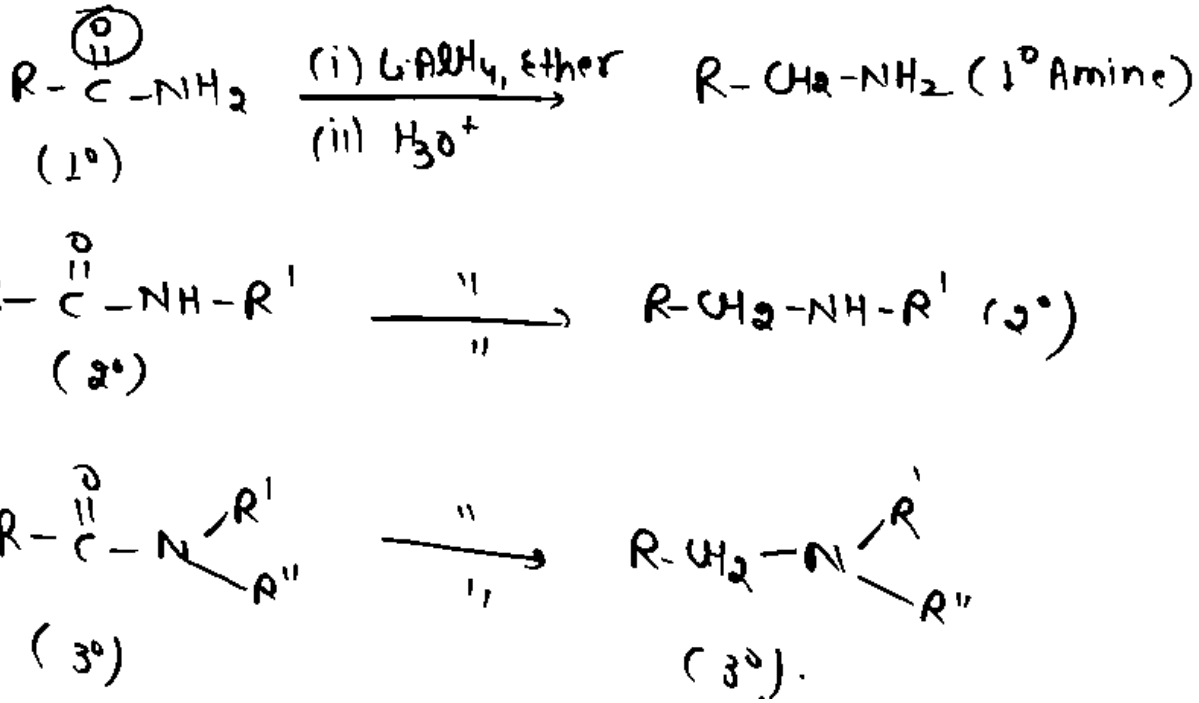
3. REDUCTION OF NITRILES (CYANIDES)
Nitriles (R-CN) are reduced to primary amines containing the same number of carbon atoms. The -CN group is converted into a $-CH_2NH_2$ group.

REAGENTS
* Lithium Aluminium Hydride: $LiAlH_4$ in ether.
* Catalytic Hydrogenation: H_2/Ni, H_2/Pt, or H_2/Pd.
* Mendius Reaction: Sodium amalgam [Na(Hg)] in Ethanol (C_2H_5OH).

- This method is particularly useful for ascending the amine series (increasing chain length) if starting from an alkyl halide via cyanide substitution.
- Can only make 1° amines.

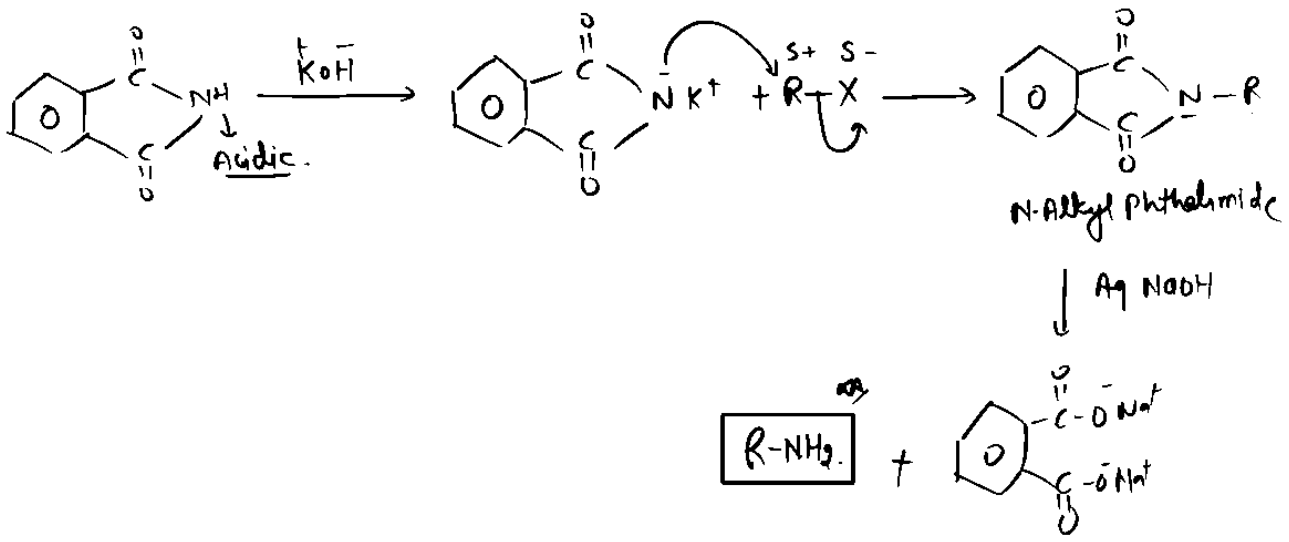
4. REDUCTION OF AMIDES
Amides ($R-CONH_2$) are reduced to amines with the same number of carbon atoms. The carbonyl group (C=O) is converted into a methylene group (CH_2).
Mechanism trick: Remove the oxygen and add two hydrogens to that carbon.

REAGENTS
* Reagent: $LiAlH_4$ followed by acidification H_3O^+/H^+.



SPECIFIC EXAMPLES
* Acetamide (CH_3CONH_2) → Ethylamine ($CH_3CH_2NH_2$)
* Benzamide ($C_6H_5CONH_2$) → Benzylamine ($C_6H_5CH_2NH_2$).
Note: Unlike other methods, this can produce 1°, 2°, or 3° amines depending on the degree of the starting amide.

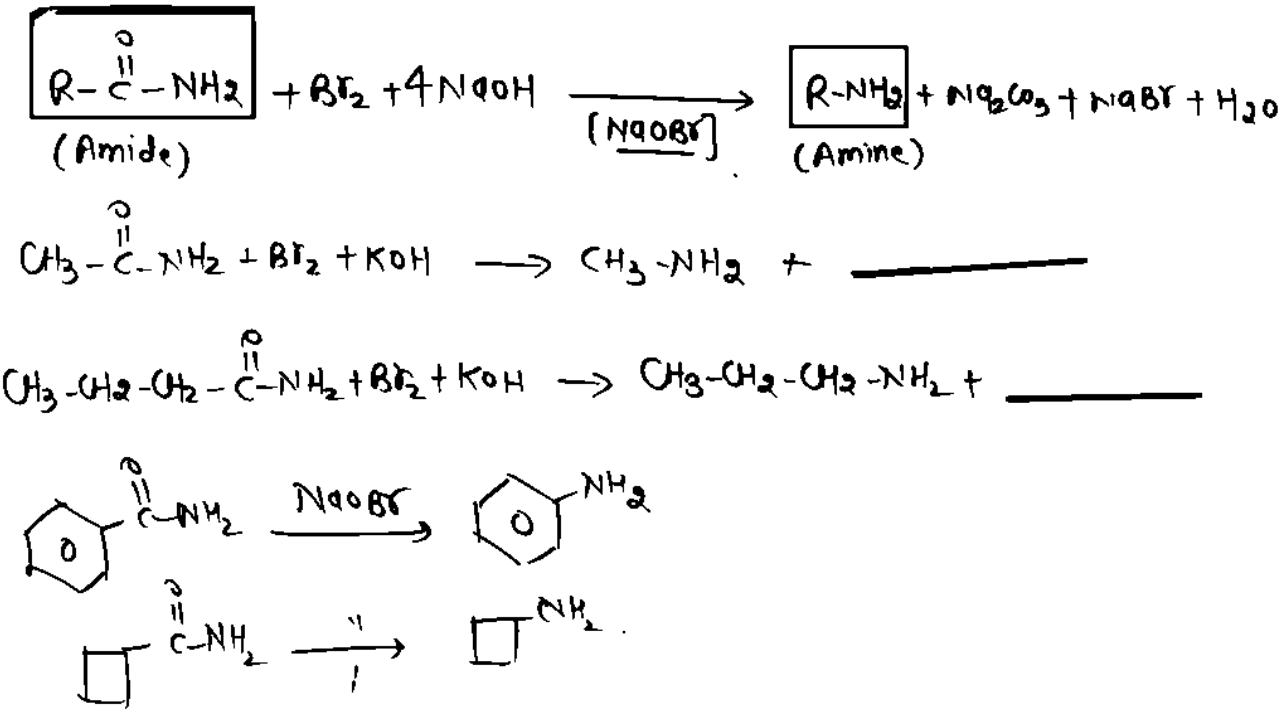
5. GABRIEL PHTHALIMIDE SYNTHESIS {V.V.Imp}
This is a specific method used for the preparation of pure 1° aliphatic amines.
Steps:
1. Phthalimide + KOH → Potassium Phthalimide (Salt).
2. Salt + R-X → N-Alkyl Phthalimide (Nucleophilic Substitution).
3. Hydrolysis with NaOH → 1° Amine ($R-NH_2$) + Sodium Phthalate.



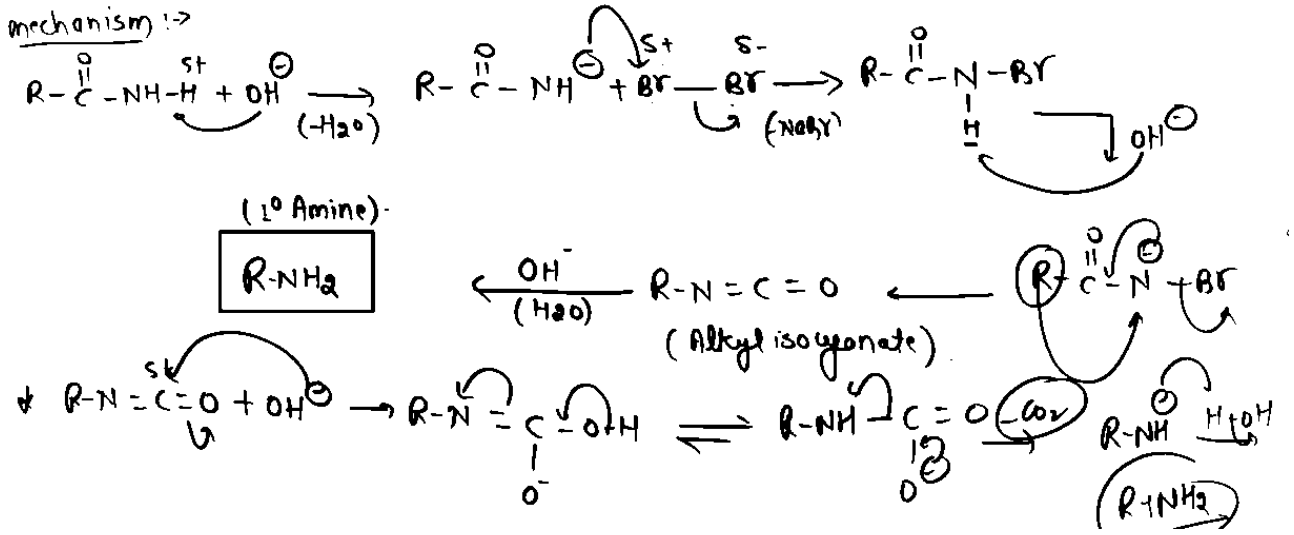
SPECIFICITY AND LIMITATIONS
* Only for 1° Aliphatic Amines: The step involving R-X is a nucleophilic substitution, if an 3° aliphatic halide is used elimination would occur in place of nucleophilic substitution.
* Aromatic Amine Limitation: Aromatic primary amines (like aniline) cannot be prepared by this method. This is because aryl halides (like Chlorobenzene) are inert toward nucleophilic substitution and will not react with the phthalimide salt.
sirf aliphatic amines bana sakte h usme bhi sirf primary amine.

6. HOFFMANN BROMAMIDE DEGRADATION REACTION {V.V.V.V.V.Imp}
This reaction converts an amide ($R\text{-}CONH_2$) into a primary amine ($R\text{-}NH_2$) having one carbon atom less than the starting amide. It is a "degradation" because the carbon chain length is reduced.

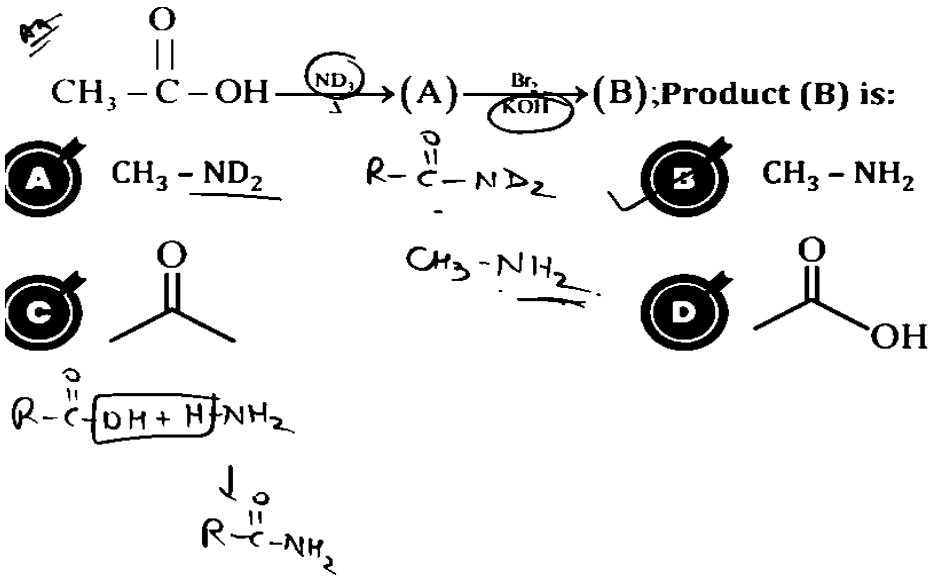
REAGENTS
* Reagent: Br_2 + NaOH/KOH **OR** *agar examiner jyada shara hoga toh* NaOBr/KOBr.
* Active species: Sodium Hypobromite (NaOBr).
* NaOH Stoichiometry: 4 moles of NaOH per mole of Amide.



MECHANISM

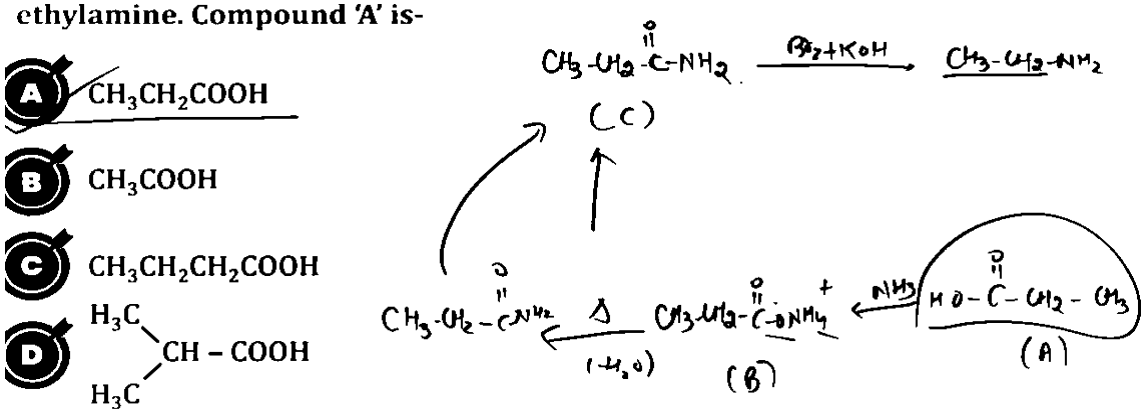


Intermediates formed -
1. N-bromoamide (R-CONHBr).
2. Alkyl Isocyanate (R-N=C=O).



N k sath jo 2 hydrogen h vo solvent k h.

An organic compound 'A' on treatment with NH_3 gives 'B', which on heating gives 'C'. 'C', when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is-

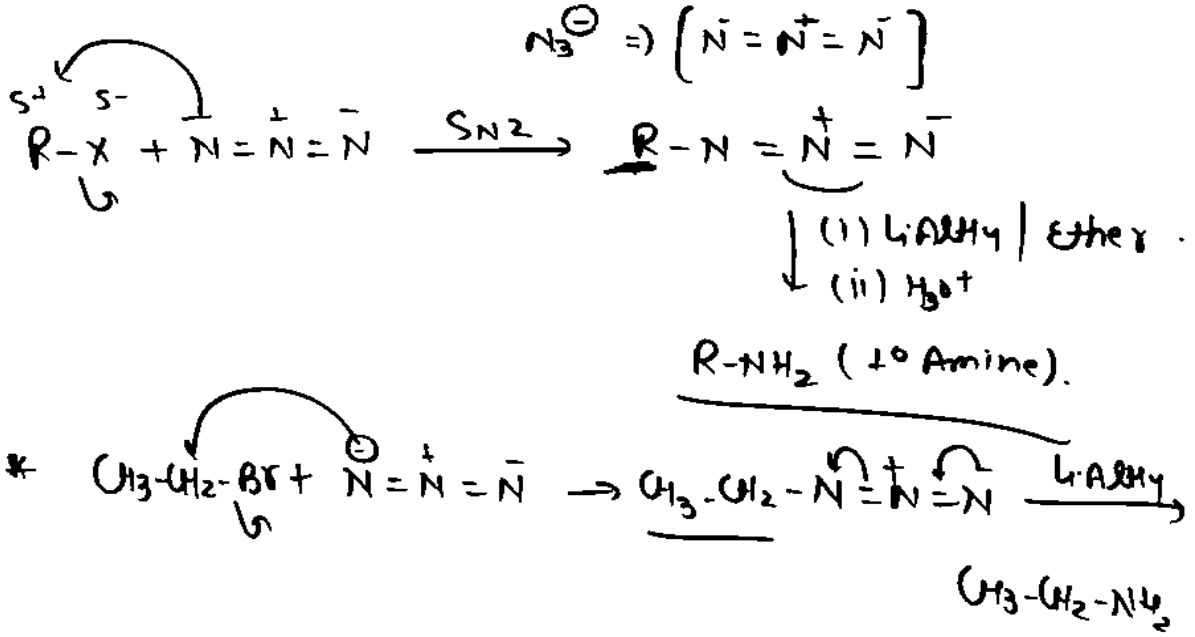


...[Iske aage ki preparation methods not so important]...

7. REDUCTION OF ISOCYANIDES (CARBYLAMINES)
- Reduction of Alkyl Isocyanides (R-NC) results in the formation of Secondary (2°) Amines.
  $R\text{-}NC + 4[H] \rightarrow R\text{-}NH\text{-}CH_3$
- Reducing agents used can be $LiAlH_4$ or H_2 with Pt/Pd/Ni
- The product is always a methyl-substituted secondary amine.

8. REDUCTION OF OXIMES
- Oximes (R-CH=N-OH) can be reduced to primary amines (1°) using Na/C_2H_5OH or $LiAlH_4$.
  $R\text{-}CH{=}N\text{-}OH \rightarrow R\text{-}CH_2\text{-}NH_2$.

9. REACTION OF AZIDES
Alkyl halides react with Sodium Azide (NaN_3) to form Alkyl Azides ($R\text{-}N_3$) via SN2 mechanism. Reduction of these azides with $LiAlH_4$ or H_2/Pd yields primary amines with the release of N_2 gas.
  $R\text{-}X + N_3^- \rightarrow R\text{-}N_3 \rightarrow R\text{-}NH_2 + N_2$



PROBLEM-SOLVING FRAMEWORK

Question Type	Identification Clues	Strategy / Approach
Carbon Loss	Amide + Br_2/NaOH	Hoffmann Degradation; remove >C=O.
Chain Ascent	R-X to $R\text{-}CH_2NH_2$	$R\text{-}X \rightarrow R\text{-}CN \rightarrow R\text{-}CH_2NH_2$.
Pure 1° Amine	Phthalimide reagent	Gabriel Synthesis; product must be aliphatic.
2° Amine Prep	Isocyanide (R-NC)	Reduction yields $R\text{-}NH\text{-}CH_3$.
Selective Nitro	Dinitro compound	Use $(NH_4)_2S$ to reduce only one NO_2.

HIGH-YIELD REVISION SHEET
MUST REMEMBER:
* Fe/HCl is best for Nitro reduction due to HCl regeneration.
* Hoffmann Degradation: Amide to Amine with ONE LESS CARBON.
* Gabriel Synthesis: NO Aromatic amines; ONLY 1° Aliphatic.
* Mendius Reaction: Na(Hg) + Ethanol for Nitrile reduction.
* Alkyl Isocyanate (R-N=C=O) is the key intermediate in Hoffmann Bromamide reaction.

COMMON TRAPS:
* Gabriel Synthesis: Aryl halides (like C_6H_5Cl) will NOT react; don't pick them for synthesis.
* Ammonolysis: Excess R-X gives a mixture; Excess NH_3 gives 1° Amine.
* 3° Alcohols/Alcohols with Cu: 3° alcohols yield alkenes, but in amines, 3° refers to N-substitution.
* Nitrile Reduction: Adds exactly one carbon to the original R-group.

PHYSICAL PROPERTIES

SOLUBILITY IN WATER
Lower aliphatic amines are generally soluble in water due to their ability to form hydrogen bonds with water molecules.
** The nitrogen atom acts as a hydrogen bond acceptor (via its lone pair), while the hydrogen atoms attached to nitrogen act as hydrogen bond donors.*
"Aromatic amines are insoluble in water."

ROLE OF HYDROGEN BONDING
- In primary (1°) and secondary (2°) amines, the nitrogen-attached hydrogen atoms participate in H-bonding with the oxygen atom of H_2O.
- The lone pair on the nitrogen atom of all amines (1°, 2°, and 3°) can also form H-bonds with the hydrogen atoms of H_2O.
- Lower aliphatic amines are highly soluble in water because the extent of H-bonding is significant.

IMPACT OF HYDROPHOBIC ALKYL GROUPS
- Amines contain two parts: the polar amine group ($-NH_2$, -NH-, or -N-) and the non-polar alkyl group (R).
- As the size of the alkyl group or the molecular mass of the amine increases, the hydrophobic part dominates and solubility in water decreases as the number of carbon atoms in the alkyl chain increases.
- Higher amines (with many carbon atoms) are essentially insoluble in water but remain soluble in organic solvents like alcohol, ether, or benzene.

SOLUBILITY TREND: 1° > 2° > 3°
The solubility follows this order because of the availability of hydrogen atoms for H-bonding.

BOILING POINT
- The boiling point (BP) is directly proportional to the strength of intermolecular forces.
- Primary and secondary amines associate with each other via intermolecular hydrogen bonding.
- 3° Amines: Since there are no hydrogen atoms attached to the nitrogen, they cannot form intermolecular H-bonds, instead, 3° amines are held together only by weaker dipole-dipole or Van der Waals forces, leading to much lower boiling points.

BOILING POINT TREND: 1° > 2° > 3°

COMPARISON WITH ALCOHOLS AND ALKANES
When comparing compounds of similar molecular mass:
Alcohols > Amines > Alkanes
- Alkanes: Only weak London dispersion/Van der Waals forces.
- Amines: Intermolecular H-bonding exists.
- Alcohols: Stronger intermolecular H-bonding than amines.

HIGH-YIELD REVISION SHEET
SOLUBILITY TRENDS
- Solubility decreases as the carbon chain length increases.
- Solubility ∝ 1 / Molecular Mass (for same degree).
- Solubility ∝ 1 / Hydrophobic group size.
- General Order: 1° > 2° > 3°.

BOILING POINT TRENDS
- Boiling Point ∝ Strength of Intermolecular Forces.
- General Order: 1° > 2° > 3°.
- Alcohols have higher BP than Amines due to stronger O-H polarity.
- Global Comparison: Carboxylic Acids > Alcohols > Amines > Alkanes.

CHEMICAL PROPERTIES

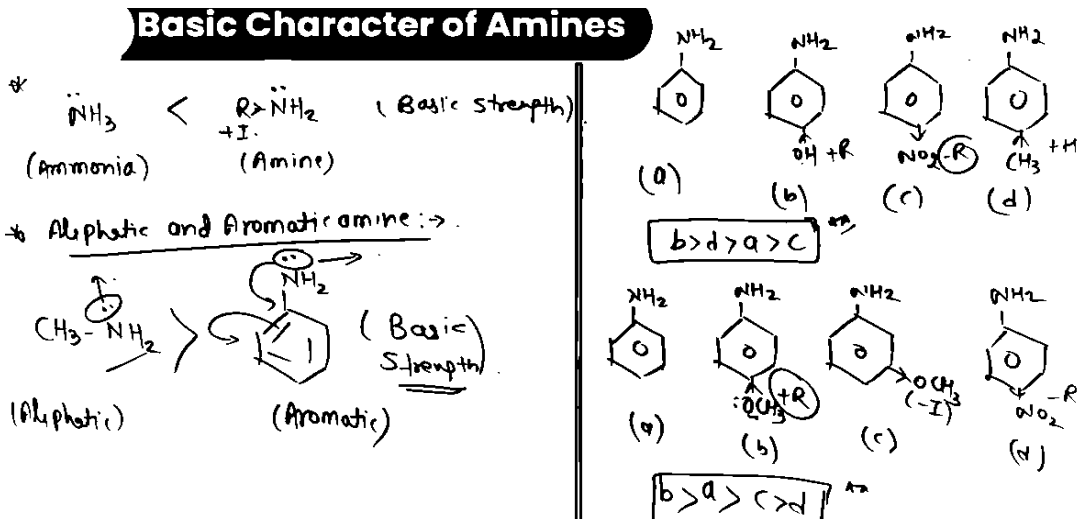
ELECTRONIC NATURE AND BASICITY

ELECTRONIC NATURE OF AMINES
Amines are reactive due to two primary electronic features:
1. The difference in electronegativity between Nitrogen and Hydrogen makes the N-H bond polar. Nitrogen carries a partial negative charge (δ^-), and Hydrogen carries a partial positive charge (δ^+).
2. The presence of an unshared non-bonding electron pair (lone pair) on the Nitrogen atom. This lone pair makes amines both nucleophilic (attacking electron-deficient centers) and basic (accepting protons).

BASIC NATURE AND SALT FORMATION {Imp}
- Amines are Lewis bases because they can donate their lone pair of electrons. -
- They react with mineral acids (like HCl, H_2SO_4) to form water-soluble quaternary ammonium salts.
- Example: $R\text{-}NH_2 + HCl \rightleftharpoons R\text{-}NH_3^+ Cl^-$ (Ammonium salt).
- These salts can be converted back to the parent amine by treatment with a strong base like NaOH.

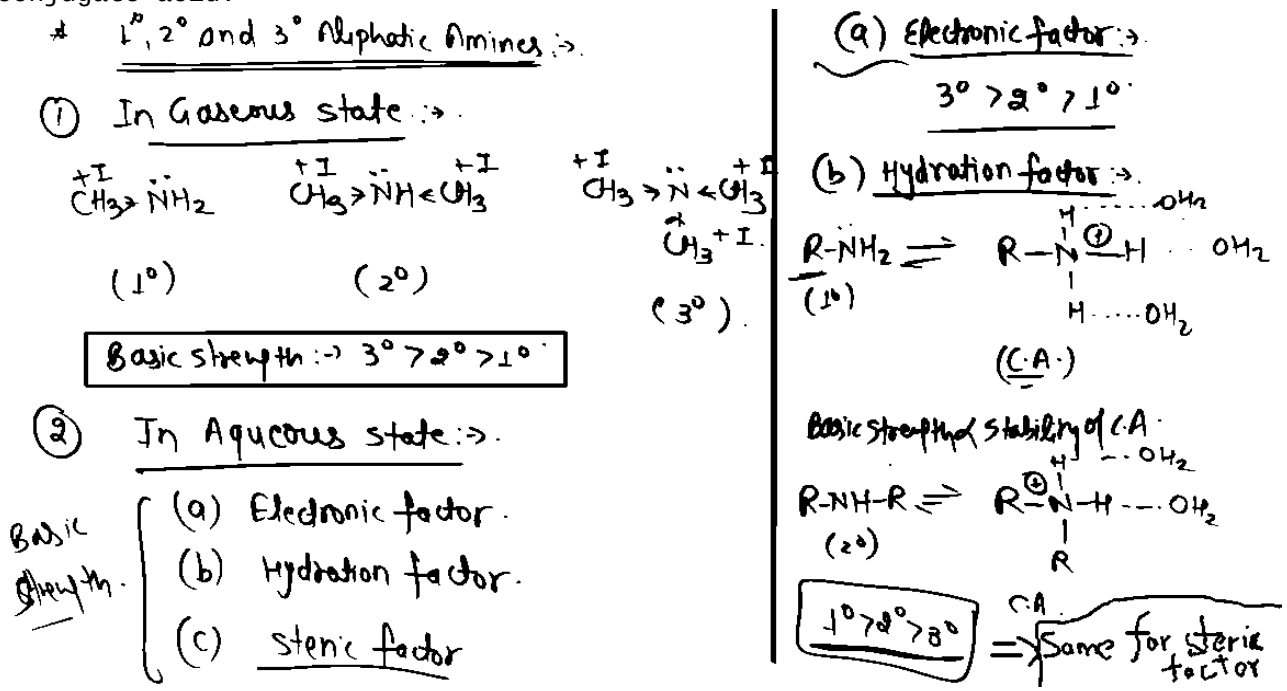
Basic Strength ∝ Stability of conjugate acid
Basic Strength ∝ Electron donating groups (EDG)
Basic Strength ∝ 1/Electron withdrawing groups (EWG)
Basic Strength↑ ⇒ K_b↑ ⇒ pK_b↓
pK_b = -log K_b.

BASICITY TRENDS
1. Aliphatic Amines vs. Ammonia: Aliphatic amines are stronger bases than ammonia due to the +I effect of alkyl groups.
2. Aliphatic vs. Aromatic Amines: Aliphatic amines are much stronger bases than aromatic amines (like Aniline). In aromatic amines, the Nitrogen lone pair is delocalized into the benzene ring via resonance, making it less available for protonation.



FACTORS AFFECTING BASICITY

1. Inductive Effect (+I): Alkyl groups (R) are electron-donating. They increase electron density on Nitrogen and stabilize the resulting conjugate acid ($R\text{-}NH_3^+$) by dispersing the positive charge.
2. Solvation/Hydration Effect: In aqueous solution, the conjugate acid is stabilized by hydrogen bonding with water molecules. The more Hydrogen atoms attached to the Nitrogen in the conjugate acid, the greater the solvation.
3. Steric Hindrance(SIR effect): Bulky alkyl groups can obstruct the approach of a proton to the Nitrogen atom and interfere with the hydration of the resulting conjugate acid.



gaseous state m order 3°>2°>1°
aur aq. state m -
2°>1°>3° if R = methyl
2°>3°>1° if R = ethyl
yaad hi karna padega :) 213 aur 231
kuch nahi bola toh gas manna h

FORMULA SHEET

RELATIONSHIP SUMMARY
- Basic Strength ∝ K_b
- Basic Strength ∝ 1 / pK_b
- Basic Strength ∝ Stability of Conjugate Acid
- pK_b = -log K_b

GAS PHASE BASICITY ORDER (PURE INDUCTIVE EFFECT)
3° Amine > 2° Amine > 1° Amine > NH_3

AQUEOUS PHASE BASICITY ORDERS
- If R = Methyl (CH_3): 2° > 1° > 3° > NH_3 (Order: 213)
- If R = Ethyl (C_2H_5): 2° > 3° > 1° > NH_3 (Order: 231)

CONCEPT CONNECTIONS

Comparison	Basic Strength	Reasoning
$R\text{-}NH_2$ vs NH_3	$R\text{-}NH_2$ > NH_3	+I effect of R-group
Aliphatic vs Aromatic	Aliphatic > Aromatic	Resonance delocalization in aromatic
Aniline vs Benzylamine	Benzylamine > Aniline	Benzylamine's LP is not in resonance

JEE ADVANCED INSIGHTS

ORTHO EFFECT AND SIP (STERIC INHIBITION OF PROTONATION)

In ortho-substituted anilines, the basicity is significantly lower than in aniline, regardless of whether the group is EDG or EWG.

- Reason: When the Nitrogen is protonated to form $Ar\text{-}NH_3^+$, the bulky ortho group creates steric repulsion with the incoming protons/water of hydration.
- Result: The conjugate acid becomes unstable, making the ortho-substituted amine a very weak base.

SIR (STERIC INHIBITION OF RESONANCE)

Observed in N,N-Dimethylaniline with bulky ortho substituents.

- Normal Case: Nitrogen lone pair is in resonance with the ring (Low basicity).
- SIR Case: Bulky ortho groups push the $-N(CH_3)_2$ group out of the ring's plane.
- Result: Resonance is blocked. The lone pair stays localized on Nitrogen, making it a MUCH stronger base (approaching aliphatic levels).

HIGH-YIELD REVISION SHEET

MUST REMEMBER

- Gas Phase Order: 3° > 2° > 1° > NH_3 (Inductive effect only).
- Aqueous Methyl: 2° > 1° > 3° (213).
- Aqueous Ethyl: 2° > 3° > 1° (231).
- Aliphatic > NH_3 > Aromatic.
- EDG increases basicity; EWG decreases basicity.
- $pK_b = -\log K_b$ (Smaller pK_b = Stronger base).

LAST-MINUTE REVISION POINTS

- SIP (Ortho Effect) makes ortho-substituted anilines weak bases.
- SIR (Steric Inhibition of Resonance) makes 3° aromatic amines with ortho groups strong bases by localized LP.
- 1° aliphatic amines + Nitrous Acid (HNO_2) → Alcohols + N_2 gas.
- Aniline + HNO_2 (0-5°C) → Stable Diazonium Salt.
- Friedel-Crafts reaction fails for Aniline because the Lewis acid ($AlCl_3$) forms a salt with the basic Nitrogen.

CHEMICAL PROPERTIES

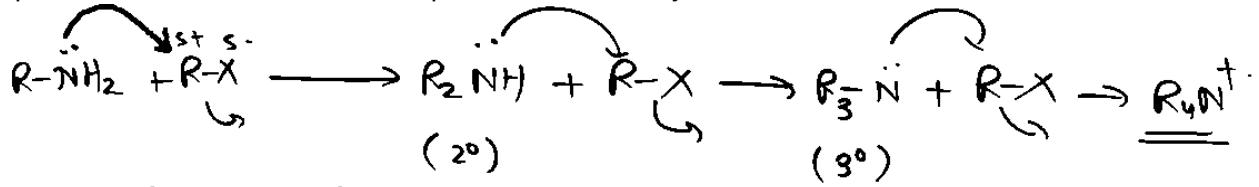
NUCLEOPHILIC REACTIONS OF AMINES

ALKYLATION OF AMINES

Amines act as nucleophiles due to the presence of a lone pair of electrons on the nitrogen atom. They react with alkyl halides (R-X) via a nucleophilic substitution (SN2) mechanism.

- In this reaction, the hydrogen atoms of the amine are successively replaced by alkyl groups.
- If the alkyl halide is in excess, the reaction proceeds from a primary (1°) amine to a secondary (2°) amine, then to a tertiary (3°) amine, and finally to a quaternary ammonium salt ($[R_4N]^+ X^-$).
- This process is also known as Hoffmann's ammonolysis of alkyl halides.

pehle kar chuke h issi chapter m ‘ammonolysis’

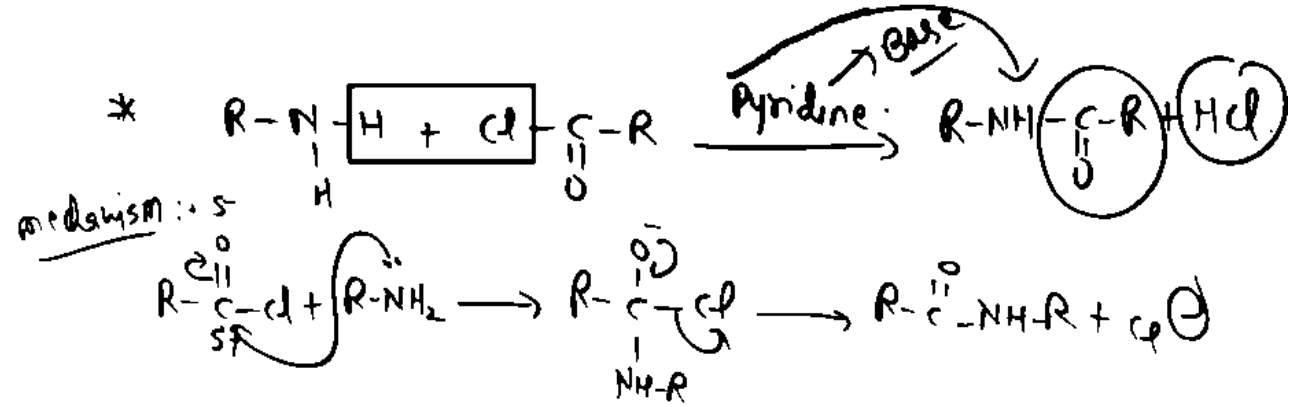


ACYLATION (ACETYLATION)

Aliphatic and aromatic primary and secondary amines react with acid chlorides, anhydrides, or esters to form amides. This reaction is a nucleophilic acyl substitution.

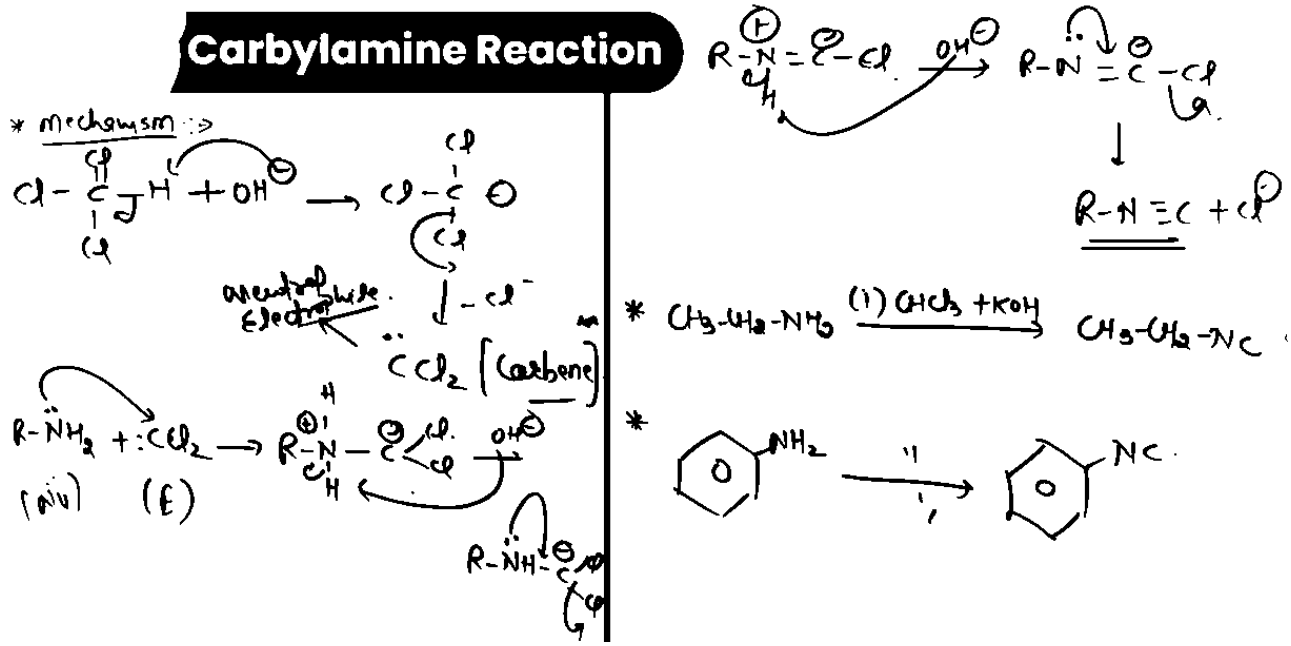
- Mechanism: Nucleophilic Addition-Elimination.
  1. The nitrogen lone pair attacks the carbonyl carbon (Addition).
  2. A tetrahedral intermediate is formed.
  3. The leaving group (e.g., Cl^- or carboxylate ion) is expelled, and a proton is lost to restore the carbonyl group (Elimination).

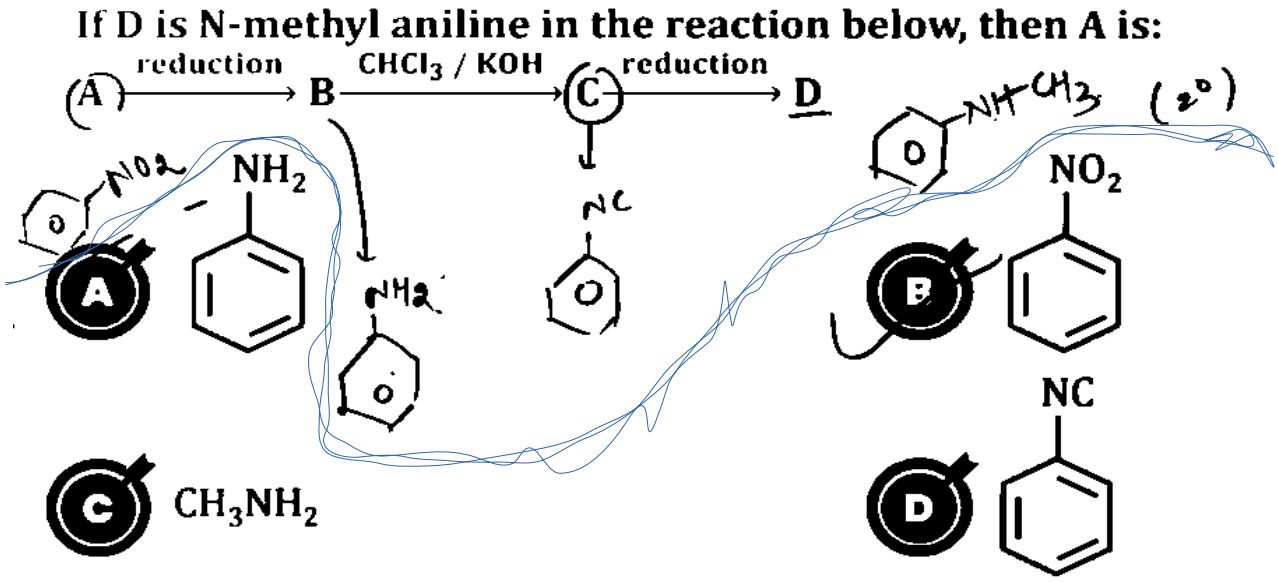
- The reaction is carried out in the presence of a base stronger than the amine, such as Pyridine.
- Purpose of Pyridine: It removes the HCl formed during the reaction, shifting the equilibrium to the right (forward direction) and preventing the amine from forming an unreactive salt with the acid.
- Note: Tertiary (3°) amines do not undergo acylation because they lack a replaceable hydrogen atom on the nitrogen.



BENZOYLATION (SCHOTTEN-BAUMANN REACTION)
When amines are reacted specifically with Benzoyl Chloride (C_6H_5COCl) in the presence of a base (like NaOH or Pyridine), the reaction is known as the Schotten-Baumann reaction.
- Example: $CH_3NH_2 + C_6H_5COCl + NaOH \rightarrow C_6H_5CONHCH_3 + NaCl + H_2O$.
same h uppar wale ke tarah :) acylation ka hi part h

CARBYLAMINE REACTION (ISOCYANIDE TEST) {Imp}
This is a specific diagnostic test used to identify primary amines.
"$R\text{-}NH_2 + CHCl_3$ --{Alc. KOH}--> $R\text{-}NC + H_2O + KCL$"
- Requirement: Only 1° aliphatic and 1° aromatic amines give this reaction.
- Reagents: Chloroform ($CHCl_3$) and Alcoholic Potassium Hydroxide (KOH).
- Product: An Isocyanide or Carbylamine (R-NC), which is characterized by an extremely offensive, foul, or unpleasant smell.
- Intermediate: The reaction proceeds via the formation of Dichlorocarbene ($:CCl_2$), which acts as a neutral electrophile.
- Note: Secondary (2°) and tertiary (3°) amines do not give this test, making it a reliable method for distinguishing primary amines from others.





INTUITION AND CONCEPTUAL UNDERSTANDING

Visualize the nitrogen atom as a ship's dock. The lone pair is the "crane" that reaches out to grab "cargo" (alkyl or acyl groups).

- In Alkylation: The crane keeps grabbing more alkyl groups as long as there is "space" (hydrogens to replace). Even when the space is full (3° amine), the crane (lone pair) is so strong it can grab one last group to form a salt.
- In Acylation: The crane grabs a bulky "shield" (the acyl group). Once this shield is on, the nitrogen is less nucleophilic because the lone pair is now busy in resonance with the new carbonyl group.
- In Carbylamine: It is like a chemical "fire alarm." Only 1° amines have the specific structure to create the "foul-smelling" isocyanide molecule. If you smell something terrible after adding $CHCl_3$ and KOH, your "1° Amine alarm" is going off.

FORMULA SHEET

ALKYLATION SEQUENCE

$R\text{-}NH_2 + R'\text{-}X \rightarrow R\text{-}NH\text{-}R' + HX$ (2° Amine)

$R\text{-}NH\text{-}R' + R'\text{-}X \rightarrow R\text{-}N(R')_2 + HX$ (3° Amine)

$R\text{-}N(R')_2 + R'\text{-}X \rightarrow [R\text{-}N(R')_3]^+ X^-$ (Quaternary Salt)

ACYLATION REACTION

$R\text{-}NH_2 + R'COCl$ (Pyridine) $\rightarrow R\text{-}NH\text{-}COR' + HCl$

$R\text{-}NH_2 + (CH_3CO)_2O \rightarrow R\text{-}NH\text{-}COCH_3 + CH_3COOH$

CARBYLAMINE REACTION

$R\text{-}NH_2 + CHCl_3 + 3KOH$ (alc.) --heat--> $R\text{-}NC + 3KCl + 3H_2O$

(Product R-NC = Alkyl Isocyanide / Carbylamine)

CONCEPT CONNECTIONS

Reaction	Linked Concept / Topic	Significance
Acylation	Protection of Amines	Used in EAS to reduce the activating power of $-NH_2$.
Carbylamine	Reimer-Tiemann Reaction	Both share the $:CCl_2$ (Dichlorocarbene) intermediate.
Alkylation	SN2 Mechanism	Follows reactivity order of halides: RI > RBr > RCl.
Acylation	Nucleophilic Acyl Sub.	Reactivity: Acid Chloride > Anhydride > Ester.

JEE ADVANCED INSIGHTS

- DICHLOROCARBENE INTERMEDIATE

In the Carbylamine reaction, KOH reacts with $CHCl_3$ to produce $:CCl_2$ via alpha-elimination. This $:CCl_2$ is an electron-deficient species (electrophile) that is attacked by the nucleophilic nitrogen of the 1° amine.

- STERIC HINDRANCE IN ACYLATION

While 1° and 2° amines react, the rate of acylation decreases significantly as the groups on the nitrogen or the acylating agent become bulkier. This is a typical Nucleophilic Addition-Elimination trend where the tetrahedral intermediate becomes harder to form.

- SELECTIVE ACYLATION

If a molecule has both an -OH and an $-NH_2$ group, the $-NH_2$ group is typically acylated first because nitrogen is more nucleophilic than oxygen.

HIGH-YIELD REVISION SHEET

MUST REMEMBER:

- Carbylamine Test = 1° Amines only + $CHCl_3$ + KOH → R-NC (Foul smell).
- Intermediate in Carbylamine = Dichlorocarbene ($:CCl_2$).
- Acylation uses Pyridine to "clean up" HCl.
- 3° Amines = No Acylation.
- Alkylation with excess R-X leads to Quaternary Ammonium Salts.
- Benzoylation name = Schotten-Baumann Reaction.

LAST-MINUTE REVISION POINTS:

- R-NC is the product of Carbylamine, NOT R-CN.
- Isocyanide smell is described as "offensive" or "foul."
- Acylation is a Nucleophilic Addition-Elimination reaction.
- Pyridine is a base that is stronger than the reactant amine.
- 1° Aromatic amines (Aniline) DO give the Carbylamine test.

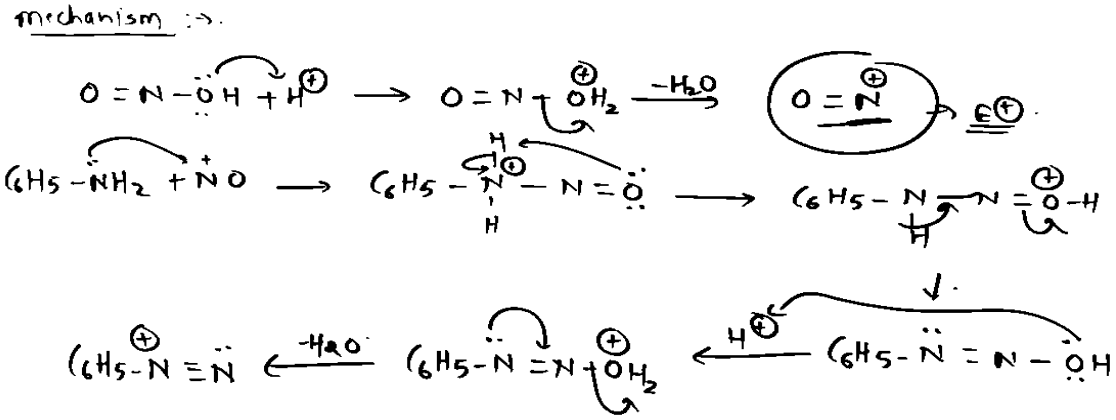
REACTIONS WITH NITROUS ACID(HNO_2) {Diazotization} {Imp}

REAGENT PREPARATION

- Nitrous acid (HNO_2) is highly unstable and cannot be stored. It must be prepared in-situ (freshly during the reaction).
- Reagent: Sodium Nitrite ($NaNO_2$) and concentrated Hydrochloric Acid (HCl).
- Reaction: $NaNO_2$ + HCl -> HNO_2 + NaCl
- Temperature: Generally maintained at 0 to 5 degrees Celsius (273 to 278 K) to prevent decomposition.
- Active Electrophile: Nitrosonium ion (NO^+), generated from HNO_2 in acidic medium.

DIAZOTIZATION OF PRIMARY AROMATIC AMINES

- Primary aromatic amines (like Aniline) react with nitrous acid at low temperatures to form stable arenediazonium salts.
- Reaction: $C_6H_5NH_2$ + $NaNO_2$ + 2HCl (at 0-5°C) -> $C_6H_5N_2^+$ Cl^- + NaCl + $2H_2O$
- Product: Benzene Diazonium Chloride.
- Stability: These salts are stable in the 0-5°C range because the diazonium group ($-N_2^+$) is stabilized by resonance with the benzene ring. If the temperature rises above 5°C, they decompose to form Phenol and release N_2 gas.

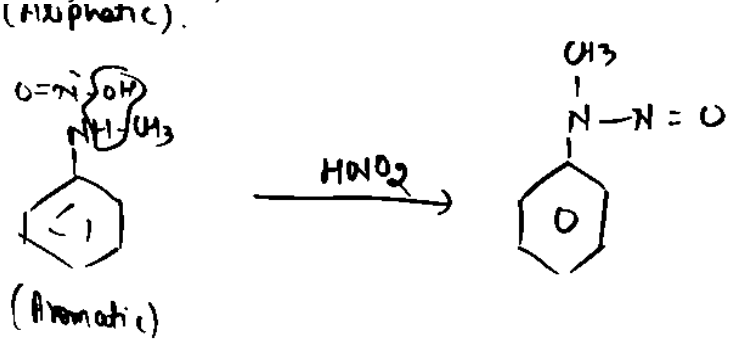


REACTION OF PRIMARY ALIPHATIC AMINES

- Primary aliphatic amines react with nitrous acid to form highly unstable aliphatic diazonium salts.
- Reaction: $R\text{-}NH_2 + HNO_2 \rightarrow [R\text{-}N_2^+ Cl^-]$ (unstable)
- Process: The unstable salt immediately decomposes to release Nitrogen gas (N_2) and forms a carbocation (R^+).
- Outcome: The carbocation typically reacts with water (from the medium) to form an Alcohol.
- Important: Since a carbocation is formed as an intermediate, rearrangement (1,2-hydride or methyl shift) can occur to form a more stable product.
- Reaction Trend: $R\text{-}NH_2 + HNO_2 \rightarrow R\text{-}OH + N_2 + H_2O$

REACTION OF SECONDARY AMINES (ALIPHATIC AND AROMATIC)

- Secondary amines react with nitrous acid to form N-nitrosoamines.
- Reaction: $R_2NH + HNO_2 \rightarrow R_2N\text{-}N{=}O + H_2O$
- Observation: N-nitrosoamines are insoluble in the reaction medium and appear as a Yellow Oily Layer.
- This reaction is characteristic of both aliphatic and aromatic secondary amines (e.g., N-methylaniline).



HNO2 ko aise rakhko ke pani nikale ::) hamesha.

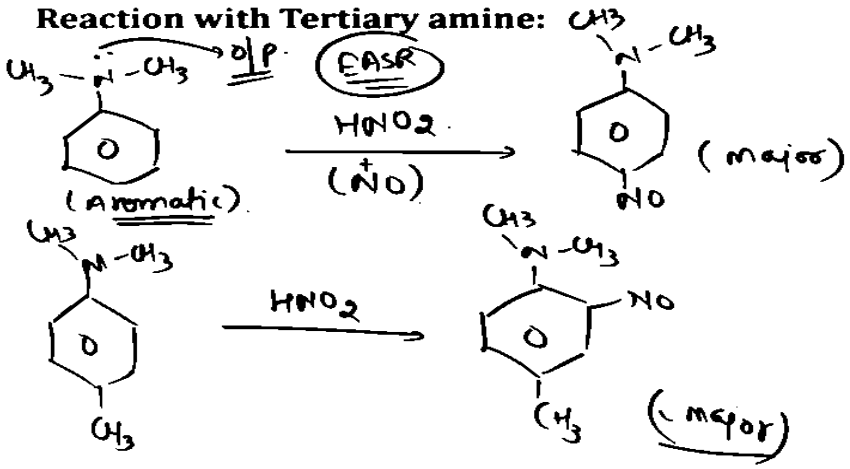
REACTION OF TERTIARY AMINES

1. ALIPHATIC TERTIARY AMINES

- Tertiary aliphatic amines lack a replaceable hydrogen on the nitrogen. They react with nitrous acid to form soluble trialkylammonium nitrite salts.
- Reaction: $R_3N + HNO_2 \rightarrow [R_3NH]^+ NO_2^-$
- This is a simple acid-base reaction forming a water-soluble salt.

2. AROMATIC TERTIARY AMINES (E.G., N,N-DIMETHYLANILINE)

- Aromatic tertiary amines undergo Electrophilic Aromatic Substitution (EAS) because the nitrogen lone pair is involved in resonance, activating the ring.
- Mechanism: The Nitrosonium ion (NO^+) attacks the para position of the aromatic ring.
- Product: Para-nitroso product (e.g., p-Nitroso-N,N-dimethylaniline).
- Note: If the para position is blocked, the substitution occurs at the ortho position, though it is less favorable due to steric hindrance.



INTUITION AND CONCEPTUAL UNDERSTANDING

THE RESONANCE STABILIZATION DIVIDE
Think of the $-N_2^+$ group as a "ticking time bomb" that wants to release N_2 gas.
- In Aliphatic Amines: There is nothing to hold the bomb. It explodes immediately, leaving a "crater" (carbocation). This explains why we see N_2 bubbles and form alcohols.
- In Aromatic Amines: The benzene ring "holds" the group through resonance. This "grip" is only strong enough at freezing temperatures (0-5°C). As soon as the temperature rises, the ring loses its grip, and the bomb goes off (releasing N_2 and forming Phenol).

VISUALIZING THE YELLOW OILY LAYER
For 2° amines, the reaction stops halfway because the nitrogen has no more hydrogens to lose. The resulting N-nitroso compound is non-polar compared to the salt/medium, which is why it separates out as a distinct yellow oil.

FORMULA SHEET

1° ALIPHATIC: $R\text{-}NH_2 + HNO_2 \rightarrow R\text{-}OH + N_2\ (gas) + H_2O$
1° AROMATIC: $C_6H_5NH_2 + HNO_2 + HCl\ (0\text{-}5°C) \rightarrow C_6H_5N_2^+\ Cl^- + 2H_2O$
2° (ALIPHATIC/AROMATIC): $R_2NH + HNO_2 \rightarrow R_2N\text{-}NO\ (Yellow\ Oil) + H_2O$
3° ALIPHATIC: $R_3N + HNO_2 \rightarrow [R_3NH]^+\ NO_2^-\ (Soluble\ Salt)$
3° AROMATIC: $Ar\text{-}NR_2 + HNO_2 \rightarrow O{=}N\text{-}Ar\text{-}NR_2$ (p-nitroso substitution)

CONCEPT CONNECTIONS

Amine Type	Immediate Product	Visual Observation	Link to other topics
1° Aliphatic	Alcohol	N_2 gas bubbles	Carbocation Rearrangement
1° Aromatic	Diazonium Salt	Clear Solution	Sandmeyer/Coupling rxn
2° Both	Nitrosoamine	Yellow oily layer	Lieberman Nitroso Test
3° Aliphatic	Nitrite Salt	Clear Solution	Acid-Base Chemistry
3° Arom	Para-nitroso	Colored product	EAS Reactions

EXAMPLES AND APPLICATIONS

EXAMPLE 1: PROPAN-1-AMINE + HNO_2
1. Formation of $CH_3CH_2CH_2\text{-}N_2^+$ (Unstable).
2. Loss of N_2 forms 1° Carbocation: $CH_3CH_2CH_2^+$.
3. 1,2-Hydride shift forms 2° Carbocation: $CH_3\text{-}CH^+\text{-}CH_3$.
4. Product: Propan-2-ol (Major) and Propan-1-ol (Minor).

EXAMPLE 2: ANILINE AT 0°C VS 25°C
At 0°C: Benzene Diazonium Chloride ($C_6H_5N_2^+\ Cl^-$) is formed.
At 25°C: The diazonium salt decomposes. Phenol (C_6H_5OH) and N_2 gas are formed.

JEE ADVANCED INSIGHTS

THE ELECTROPHILE (NO^+)
The nitrosating species is the Nitrosonium ion (NO^+). In the reaction between a 1° amine and NO^+, the Rate Determining Step (RDS) is typically the nucleophilic attack of the amine on the NO^+ ion or the subsequent proton shifts depending on conditions.

EAS ON 3° AROMATIC AMINES
For N,N-Dimethylaniline, the Nitrosonium ion is a weak electrophile. It can only attack highly activated rings (like those with $-NH_2$, $-OH$, or $-NR_2$). This is why it specifically targets the para position of aromatic amines, providing a green-colored p-nitroso derivative.

HIGH-YIELD REVISION SHEET

MUST REMEMBER

- 1° Aliphatic -> Alcohol + N_2 bubbles (Carbocation rearrangement possible).
- 1° Aromatic -> Diazonium salt (Stable ONLY at 0-5°C).
- 2° (Any) -> Yellow Oily Layer (Nitrosoamine).
- 3° Aliphatic -> Clear soluble salt.
- 3° Aromatic -> Para-nitroso product (EAS).
- Reagent $NaNO_2$ + HCl generates HNO_2 in-situ.

LAST-MINUTE REVISION POINTS

- The evolution of N_2 gas is a diagnostic test for 1° aliphatic amines.
- Benzene diazonium salts are essential intermediates for Sandmeyer and Coupling reactions.
- Only the 2° amine gives the "Yellow Oily Layer" observation.
- If para is blocked in a 3° aromatic amine, NO^+ goes to the ortho position.

HINSBERG TEST

Hinsberg Test is a chemical method used to distinguish between primary (1°), secondary (2°), and tertiary (3°) amines. It can also be used for the separation of a mixture of these amines.

HINSBERG REAGENT

The reagent used is Benzene Sulphonyl Chloride. Its chemical formula is $C_6H_5SO_2Cl$.

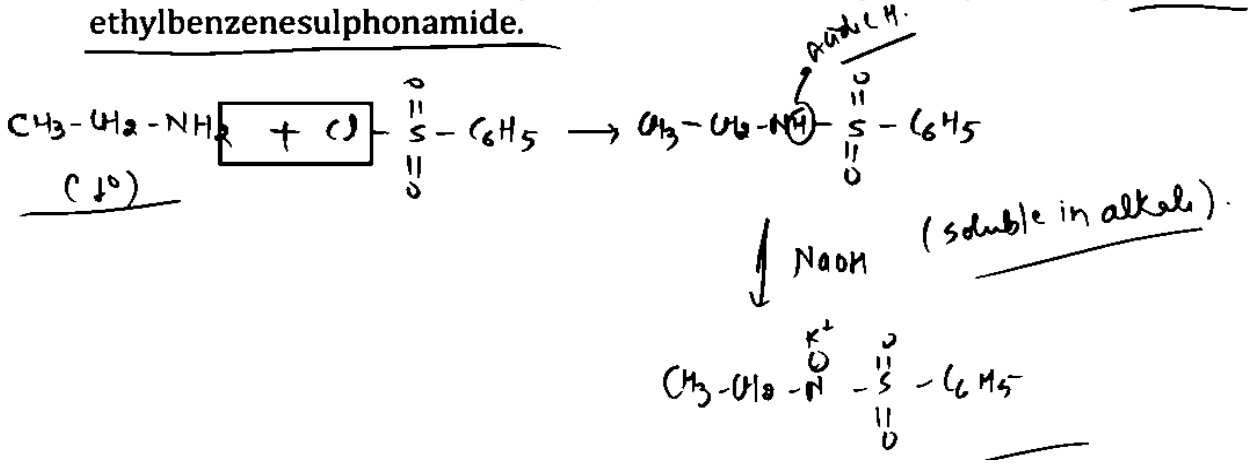
REACTION MECHANISM

The reaction is a nucleophilic substitution. The lone pair of the amine nitrogen attacks the sulfur atom of the benzene sulphonyl chloride, displacing the chloride ion (Cl^-).

REACTION WITH 1° AMINES

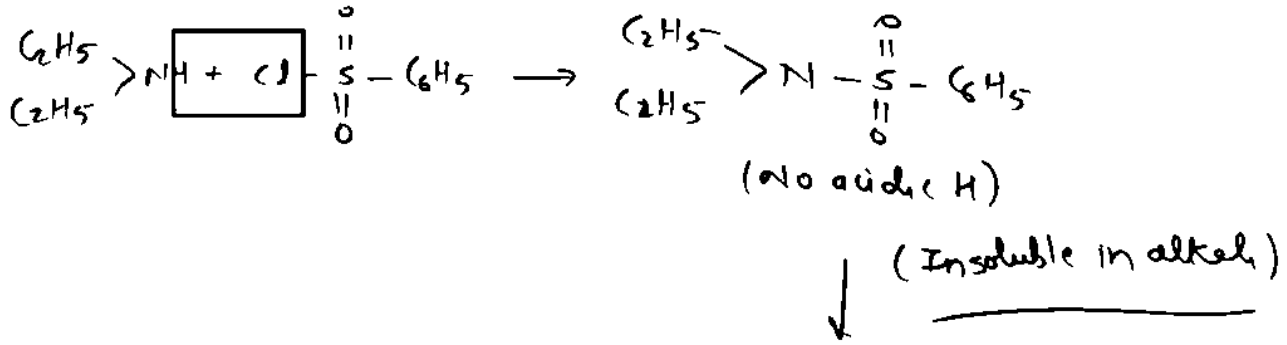
- A primary amine reacts with benzene sulphonyl chloride to form N-alkyl benzene sulphonamide.
- The resulting product contains one acidic hydrogen atom attached directly to the nitrogen atom.
- The strong electron-withdrawing nature of the sulphonyl group (-SO_2-) makes this hydrogen highly acidic.
- Because of this acidic hydrogen, the product is soluble in alkali (like NaOH or KOH).

(a) The reaction of benzenesulphonyl chloride with primary amine yields N-ethylbenzenesulphonamide.



REACTION WITH 2° AMINES

- A secondary amine reacts with benzene sulphonyl chloride to form N,N-dialkyl benzene sulphonamide.
- The resulting product has no hydrogen atom attached to the nitrogen atom.
- Therefore, the product is not acidic and is insoluble in alkali. It usually appears as a precipitate.



REACTION WITH 3° AMINES

Tertiary amines do not have any replaceable hydrogen atoms on the nitrogen atom. Consequently, tertiary amines do not react with benzene sulphonyl chloride.

INTUITION AND CONCEPTUAL UNDERSTANDING

THE ACIDIC HYDROGEN KEY

- The entire test relies on whether the nitrogen atom in the final product still holds a "spare" hydrogen.
- In a 1° amine ($R\text{-}NH_2$), two hydrogens are available. One is lost to remove the
- Chlorine, and one remains on the Nitrogen. This remaining hydrogen is "squeezed" by the powerful $-SO_2-$ group, making it easy to pop off as an H^+ (acidic). This allows the molecule to dissolve in a base.
- In a 2° amine (R_2NH), only one hydrogen is available. It is used up in the reaction to remove the Chlorine. The nitrogen is left "bare" of hydrogens, so it cannot interact with a base.

FORMULA SHEET

HINSBERG REAGENT: $C_6H_5SO_2Cl$

1° AMINE REACTION:
$C_6H_5SO_2Cl + R\text{-}NH_2 \rightarrow C_6H_5SO_2NH\text{-}R + HCl$
(Product + NaOH → Soluble Salt)

2° AMINE REACTION:
$C_6H_5SO_2Cl + R_2NH \rightarrow C_6H_5SO_2N(R)_2 + HCl$
(Product + NaOH → Insoluble/No change)

3° AMINE REACTION:
$C_6H_5SO_2Cl + R_3N \rightarrow$ No Reaction

CONCEPT CONNECTIONS

Comparison Point	1° Amine	2° Amine	3° Amine
Reaction with $C_6H_5SO_2Cl$	Yes	Yes	No
Hydrogens on Product N	One (Acidic)	Zero	N/A
Behavior in NaOH	Soluble	Insoluble	N/A

JEE ADVANCED INSIGHTS

SEPARATION OF MIXTURES

This test is not just for identification but for physical separation.
If you have a mixture of 1°, 2°, and 3° amines:
1. Add Hinsberg Reagent.
2. 3° amine remains unreacted.
3. Add alkali. 1° amine product dissolves, 2° amine product stays as a precipitate.
4. Filter to remove 2° product.
5. Distill to recover 3° amine.
6. Acidify the filtrate to recover the 1° amine product.

HIGH-YIELD REVISION SHEET

MUST REMEMBER:

- Reagent: Benzene Sulphonyl Chloride ($C_6H_5SO_2Cl$).
- 1° Amine: Reacts → Soluble in NaOH.
- 2° Amine: Reacts → Insoluble in NaOH.
- 3° Amine: No Reaction.

LAST-MINUTE REVISION POINTS:

- The product of 1° amine is soluble because of the acidic H on Nitrogen.
- The $-SO_2-$ group is a strong electron-withdrawing group that creates this acidity.
- This test effectively separates or distinguishes 1°, 2°, and 3° amines.
- Tertiary amines lack a replaceable H, which is why they don't react.

ELECTROPHILIC AROMATIC SUBSTITUTION REACTION {Imp}

ACTIVATION AND DIRECTING INFLUENCE

- The $-NH_2$ group is a very strong activating group for electrophilic aromatic substitution (EAS).
- This is because the lone pair of electrons on the nitrogen atom is in direct resonance with the benzene ring, significantly increasing the electron density at the ortho and para positions.
- Nitrogen is less electronegative than Oxygen, making its lone pair more available for donation compared to the -OH group in phenol.
- Consequently, aromatic amines like aniline are highly reactive towards electrophiles and are exclusively ortho-para directing.

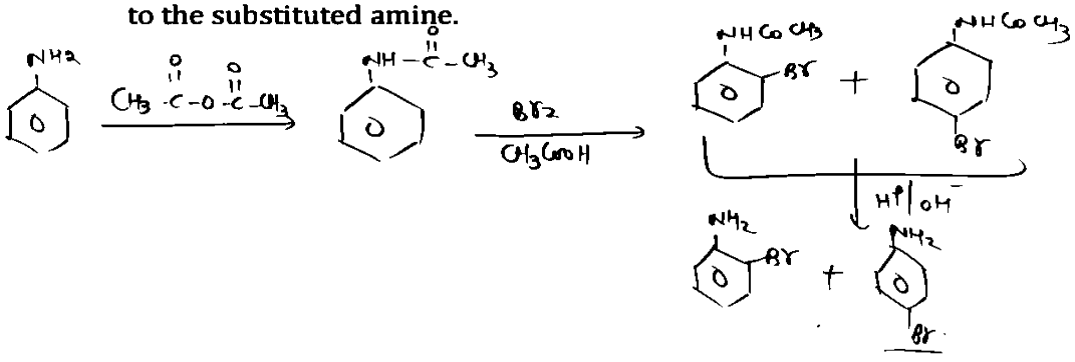
BROMINATION

- When aniline is treated with bromine water (Br_2/H_2O) at room temperature, it undergoes immediate and multiple substitutions.
- All available ortho and para positions are occupied, resulting in the formation of 2,4,6-tribromoaniline. # as in phenol
- Observation: This product appears as a distinct <u>white precipitate</u>.

<u>MONOBROMINATION STRATEGY (PROTECTION)</u>

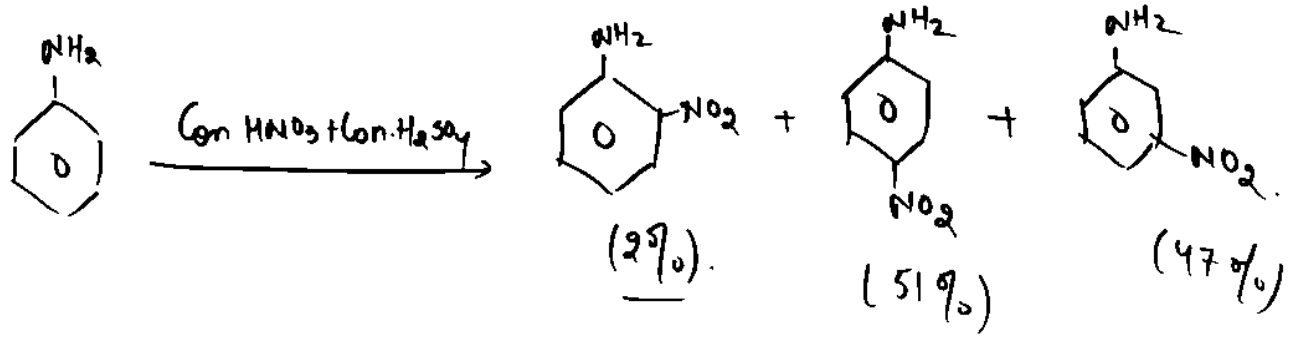
- Because the $-NH_2$ group is so reactive, it is difficult to stop the reaction at the mono-substitution stage.
- To obtain a mono-substituted product (like para-bromoaniline), the activating power of the $-NH_2$ group must be decreased.
- This is achieved through acetylation (protection) using acetic anhydride $[(CH_3CO)_2O]$ in the presence of pyridine.
- The reaction forms acetanilide ($C_6H_5NHCOCH_3$).
- In acetanilide, the nitrogen lone pair is involved in cross-conjugation with the carbonyl group, making it less available for the ring.
- After bromination of acetanilide, the acetyl group is removed via acidic or basic hydrolysis to regenerate the amine group, yielding the para-bromoaniline as the major product.

➢ **Monosubstituted aniline can be produced by protecting the $-NH_2$ group by acetylation with acetic anhydride, then carrying out the desired substitution followed by hydrolysis of the substituted amide to the substituted amine.**

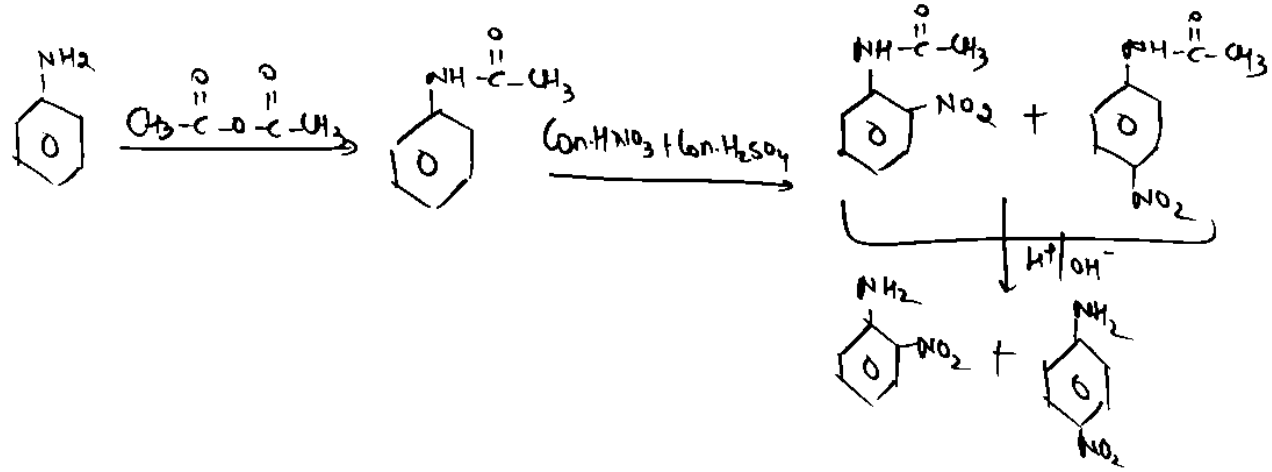


NITRATION ANOMALY

- Direct nitration of aniline with a mixture of concentrated HNO_3 and H_2SO_4 leads to a surprising result, instead of just ortho and para products, an appreciable amount of the meta product is formed.
- Composition: Para-nitroaniline (51%), Meta-nitroaniline (47%), and Ortho-nitroaniline (2%).
- Reason: In the strongly acidic nitrating medium, the basic aniline molecule is protonated to form the anilinium ion (C_6H_5-NH_3^+).
- The -NH_3^+ group is a strong electron-withdrawing group and is meta-directing.

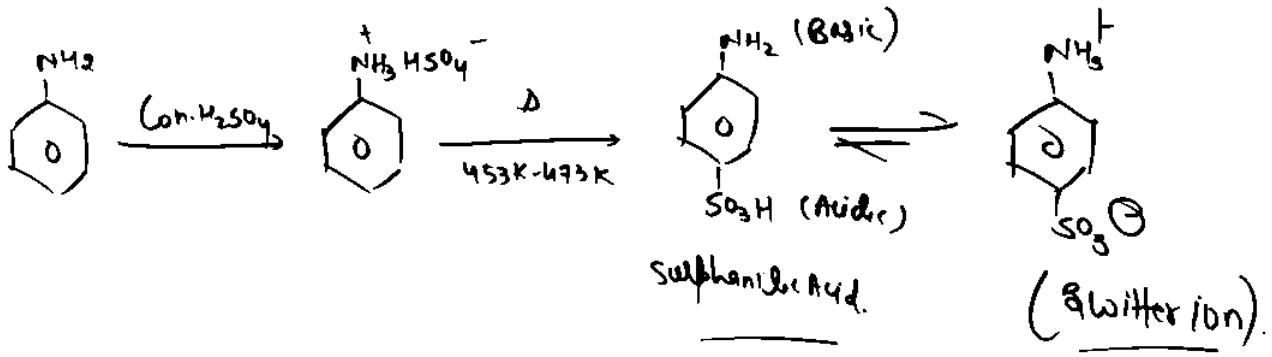


- To obtain the para product as major, aniline is first protected by acetylation, then nitrated, and finally hydrolyzed. #davai vo hi h jo pehle thi



SULPHONATION

- Aniline reacts with concentrated H_2SO_4 to initially form a salt called anilinium hydrogen sulphate.
- When this salt is heated to 453-473 K, it undergoes a rearrangement to form p-aminobenzene sulphonic acid, commonly known as sulphanilic acid.
- Sulphanilic acid contains both a basic (-NH_2) group and an acidic (-SO_3H) group within the same molecule, therefore it exists in an equilibrium as a dipolar ion known as a Zwitterion [(+NH_3)-C_6H_4-(SO_3^-)].



FRIEDEL-CRAFTS REACTION FAILURE
- Aniline and other aromatic amines do not undergo Friedel-Crafts alkylation or acylation.
- Reason: The Lewis acid catalyst used in the reaction ($AlCl_3$) is an electrophile, while the $-NH_2$ group is a Lewis base, They react immediately to form a salt.
- The nitrogen atom in the salt carries a positive charge ($C_6H_5NH_2^+$-$AlCl_3^-$), which makes it a strong deactivating group, preventing any further EAS from occurring.

FORMULA SHEET

BROMINATION
Reaction: $C_6H_5NH_2 + 3Br_2$ (aq) -> $C_6H_2Br_3NH_2$ (2,4,6-tribromoaniline) + 3HBr
Observation: White Precipitate

NITRATION PRODUCT DISTRIBUTION
- Para (Major): 51%
- Meta (Significant): 47%
- Ortho (Minor): 2%

SULPHONATION AND ZWITTERION
Aniline + H_2SO_4 (conc.) -> $C_6H_5NH_3^+ HSO_4^-$ --(453-473 K)--> NH_2-C_6H_4-SO_3H
Zwitterion: $^+NH_3$-C_6H_4-SO_3^-

PROTECTION REACTION
$C_6H_5NH_2 + (CH_3CO)_2O$ --(Pyridine)--> $C_6H_5NHCOCH_3$ (Acetanilide)

CONCEPT CONNECTIONS

Reaction Type	Reagent	Aniline Result	Reason
Bromination	Br_2/H_2O	2,4,6-Tribromo	High activation
Nitration	HNO_3/H_2SO_4	47% Meta	Anilinium ion formation
Sulphonation	H_2SO_4 (heat)	Zwitterion	Acid-Base internal reaction
Friedel-Crafts	R-Cl/$AlCl_3$	No Reaction	Salt formation with catalyst

HIGH-YIELD REVISION SHEET
MUST REMEMBER:
- Aniline + Br_2 (aq) = White ppt (2,4,6-tribromo).
- Nitration gives 47% Meta because of $-NH_3^+$.
- Protection = Acetylation (use acetic anhydride and pyridine).
- Sulphanilic acid exists as a Zwitterion.
- Friedel-Crafts fails due to salt formation with $AlCl_3$.

LAST-MINUTE REVISION POINTS:
- Temperature for Sulphonation: 453 to 473 K.
- The protective group is $NHCOCH_3$ (Acetanilide).
- Order of reactivity: Aniline > Benzene > Nitrobenzene.
- Hydrolysis is needed to "unprotect" the amine.

DIAZONIUM SALTS

- Diazonium salts are a class of organic compounds with the general formula:
  $Ar\text{-}N_2^+ X^-$.
Ar represents an aryl group (aromatic ring), and X^- is an anion such as Cl^-, Br^-, HSO_4^-, BF_4^-, etc.
- The N_2^+ group is called the diazonium group.

STABILITY AND RESONANCE

- Primary aliphatic amines form highly unstable diazonium salts that decompose immediately into carbocations and nitrogen gas.
- Primary aromatic diazonium salts are stable for a short time in aqueous solution at low temperatures (0-5°C or 273-278 K).
- Stability Reason: The stability of aryl diazonium salts is due to the dispersal of the positive charge through resonance with the benzene ring.
- Warning: Above 5°C, they decompose to form phenol and release nitrogen gas.

METHOD OF PREPARATION

PREPARATION (DIAZOTIZATION)

- Aryl diazonium salts are prepared by reacting a primary aromatic amine (like aniline) with nitrous acid (HNO_2) at 0-5°C.
- Nitrous acid is produced in the reaction mixture (in-situ) by the reaction of sodium nitrite ($NaNO_2$) with hydrochloric acid (Hcl).
- The conversion of primary aromatic amines into diazonium salts is known as diazotisation.
- Due to it's instability, the diazonium salt is not generally stored and is used immediately after it's preparation.
- Reaction: $C_6H_5NH_2 + NaNO_2 + 2HCl$ (at 273-278 K) $\rightarrow C_6H_5N_2^+ Cl^- + NaCl + 2H_2O$.

{ When performing displacement, notice that nitrogen (N_2) is one of the best leaving groups in organic chemistry because it is a very stable gas. This makes diazonium salts a "gateway" to many other benzene derivatives that cannot be prepared by direct substitution. }

PHYSICAL PROPERTIES

- Benzenediazonium chloride is a colourless crystalline solid.
- It is readily soluble in water and is stable in cold but reacts with water when warmed.
- It decomposes easily in the dry state.
- Benzenediazonium fluoroborate is water insoluble and stable at room temperature.

CHEMICAL REACTIONS

REACTIONS INVOLVING DISPLACEMENT OF N_2

1. Sandmeyer Reaction:
   - Cu_2Cl_2 / HCl $\rightarrow C_6H_5Cl + N_2$
   - Cu_2Br_2 / HBr $\rightarrow C_6H_5Br + N_2$
   - Cu_2CN_2 / KCN $\rightarrow C_6H_5CN + N_2$

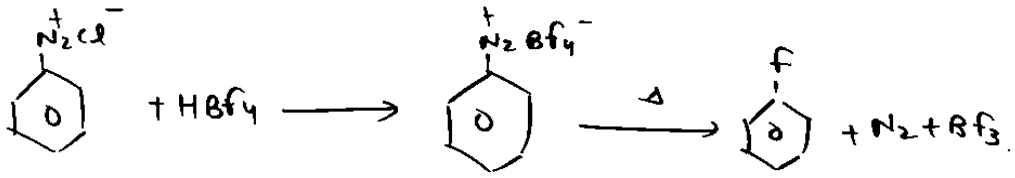
2. Gattermann Reaction:
   - Cu powder / HCl $\rightarrow C_6H_5Cl + N_2$
   - Cu powder / HBr $\rightarrow C_6H_5Br + N_2$

3. Replacement by Iodine:
   - KI / Warm $\rightarrow C_6H_5I + N_2 + KCl$

4. Balz-Schiemann Reaction (Fluorine):
   - HBF_4 (heat) $\rightarrow C_6H_5F + BF_3 + N_2$

3. Replacement by fluoride ion:

- **When arenediazonium chloride is treated with fluoroboric acid, arene diazonium fluoroborate is precipitated which on heating decomposes to yield aryl fluoride.**

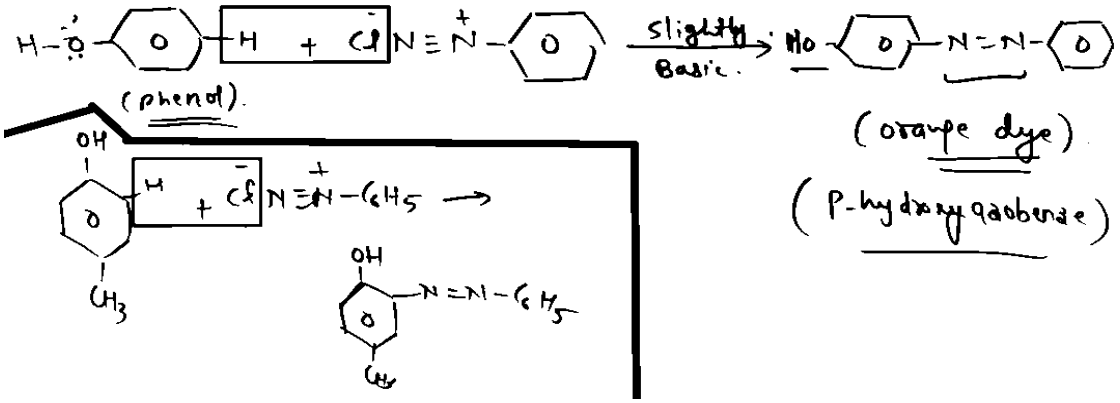


5. Reduction to Benzene:
   - $H_3PO_2 + H_2O \rightarrow C_6H_5H + H_3PO_3 + N_2$
   - $CH_3CH_2OH \rightarrow C_6H_5H + CH_3CHO + N_2$

6. Hydrolysis (Phenol):
   - H_2O (Warm/283 K) $\rightarrow C_6H_5OH + N_2 + HCl$

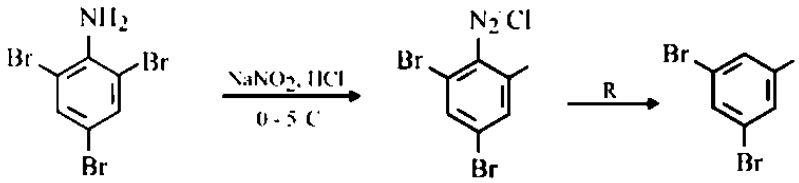
7. Replacement by NO_2:
   - HBF_4 then $NaNO_2$ / Cu (heat) $\rightarrow C_6H_5NO_2 + N_2$

REACTION INVOLVING RETENTION OF DIAZO GROUP
COUPLING REACTIONS (N_2 RETENTION)
- With Phenol (pH 9-10): $Ar\text{-}N_2^+ + C_6H_5OH \rightarrow$ p-hydroxyazobenzene (Orange Dye)
- With Aniline (pH 4-5): $Ar\text{-}N_2^+ + C_6H_5NH_2 \rightarrow$ p-aminoazobenzene (Yellow Dye)



aniline m sab same h bas -OH ki jagah NH_3 hoga

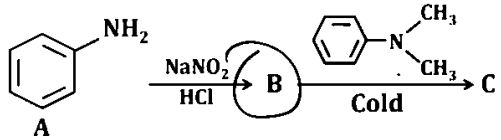
The reagent 'R' in the given sequence of a chemical reaction is:

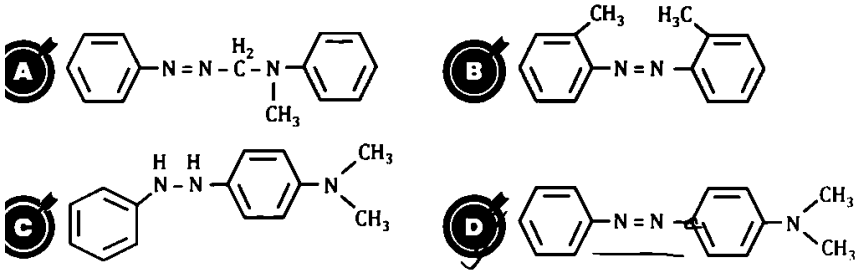


(A) HI (B) CuCN/KCN

(C) H_2O (D) CH_3CH_2OH

The structure of C in the below mentioned reaction is





CONCEPT CONNECTIONS

Property / Reaction	Connection to Other Topics
Stability	Resonance (General Organic Chemistry)
Sandmeyer	Nucleophilic Substitution (Aryl Halides)
Balz-Schiemann	Specific method for Fluoro-aromatics
Coupling	Electrophilic Aromatic Substitution (EAS)
N_2 Gas Evolution	Diagnostic test for primary amines

JEE ADVANCED INSIGHTS

ELECTROPHILIC NATURE
The diazonium ion ($Ar\text{-}N_2^+$) acts as a weak electrophile in coupling reactions. It is not strong enough to attack simple benzene; it needs a highly activated ring like Phenol or Aniline to react.

COUPLING SELECTIVITY
Coupling always takes place at the para position with respect to the activating group (-OH or $-NH_2$). If the para position is blocked, it occurs at the ortho position.

H_3PO_2 VS ETHANOL
In the reduction to benzene, H_3PO_2 is oxidized to phosphorous acid (H_3PO_3), while Ethanol is oxidized to Acetaldehyde (CH_3CHO). Both acts as reducing agents for the diazonium salt.

HIGH-YIELD REVISION SHEET

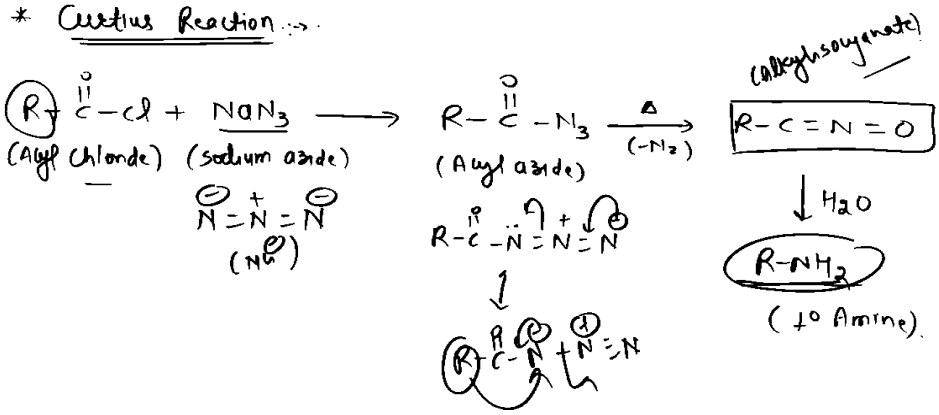
MUST REMEMBER:
- Preparation: Aniline + $NaNO_2$ + HCl (0-5°C).
- Sandmeyer: Cu(I) salts; Gattermann: Cu powder.
- Iodine: Use KI (no catalyst needed).
- Fluorine: Balz-Schiemann using HBF_4.
- Reduction: H_3PO_2 or Ethanol.
- Hydrolysis: Warm water → Phenol.
- Coupling: Diazonium + Phenol → Orange dye (Basic pH).
- Coupling: Diazonium + Aniline → Yellow dye (Acidic pH).

LAST-MINUTE REVISION POINTS:
- Aliphatic 1° amines produce N_2 gas and alcohol; they do not form stable salts.
- The $-N_2^+$ group is meta-directing and deactivating if left on the ring (before displacement).
- Diazonium salts are the best intermediates for introducing -F, -I, -CN, and -OH groups into the benzene ring.
- Coupling is an example of an Electrophilic Aromatic Substitution (EAS) reaction.

OTHER/ADDITIONAL NAME REACTIONS: CURTIUS AND SCHMIDT REACTIONS {not so imp}

Both reactions are methods to prepare primary amines (1°) from carboxylic acid derivatives. They are closely related to the Hoffmann Bromamide Degradation because they involve the same key intermediate: Alkyl Isocyanate (R-N=C=O).

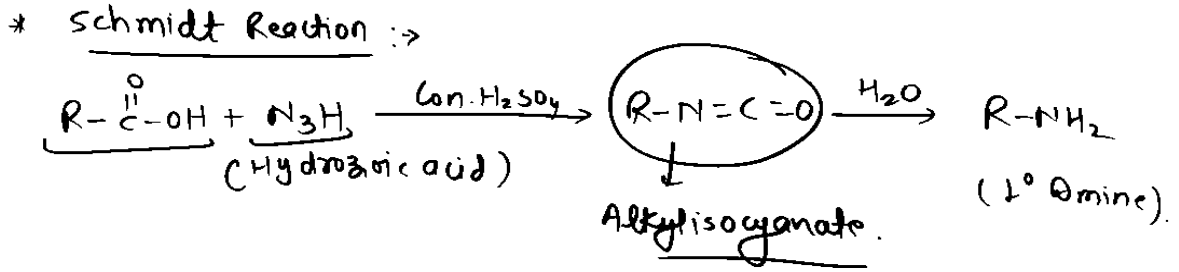
A. CURTIUS REACTION
In this reaction, an acyl chloride (R-COCl) is treated with sodium azide (NaN_3).
- Step 1: Nucleophilic substitution occurs where the azide ion (N_3^-) replaces the chloride to form an Acyl Azide ($R\text{-}CON_3$).
- Step 2: Upon heating, the acyl azide undergoes rearrangement with the loss of Nitrogen gas (N_2) to form Alkyl Isocyanate (R-N=C=O).
- Step 3: Hydrolysis of the isocyanate yields a primary amine ($R\text{-}NH_2$).



B. SCHMIDT REACTION

This involves the reaction of a carboxylic acid (R-COOH) with hydrazoic acid (N_3H) in the presence of concentrated H_2SO_4.

- Like the Curtius reaction, it proceeds via an alkyl isocyanate intermediate.
- The final product is a primary amine with one less carbon atom than the starting acid.
- (Warning) Do not confuse Sodium Azide (NaN_3) used in Curtius with Hydrazoic Acid (N_3H) used in Schmidt.



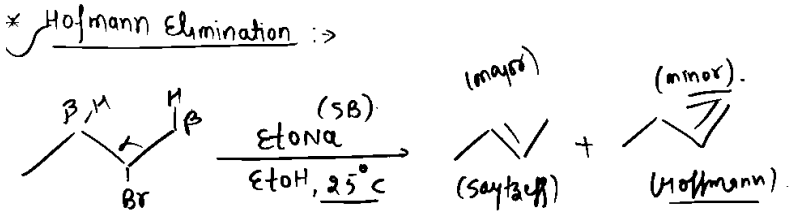
INTUITION AND CONCEPTUAL UNDERSTANDING

Think of these reactions as "carbon-cutting" tools. Just like Hoffmann Bromamide, you are removing the carbonyl carbon (C=O) and attaching the R group directly to the Nitrogen. The "engine" that drives these reactions is the extremely high stability of the N_2 gas that leaves during the rearrangement.

Curtius: $R\text{-}COCl + NaN_3$ (heat) + $H_2O \rightarrow R\text{-}NH_2 + N_2 + NaCl$
Schmidt: $R\text{-}COOH + N_3H$ (conc. H_2SO_4) $\rightarrow R\text{-}NH_2 + CO_2 + N_2$

C. HOFFMANN ELIMINATION (ANTI-SAYTZEFF/HOFMANN PRODUCT)

In standard elimination reactions (E2) of alkyl halides, the more substituted alkene (Saytzeff product) is usually the major product. However, in the elimination of quaternary ammonium salts ($R_4N^+ X^-$), the trend is reversed.

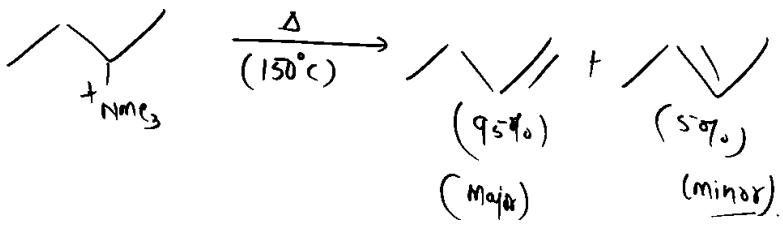


D. HOFFMANN MUSTARD OIL REACTION

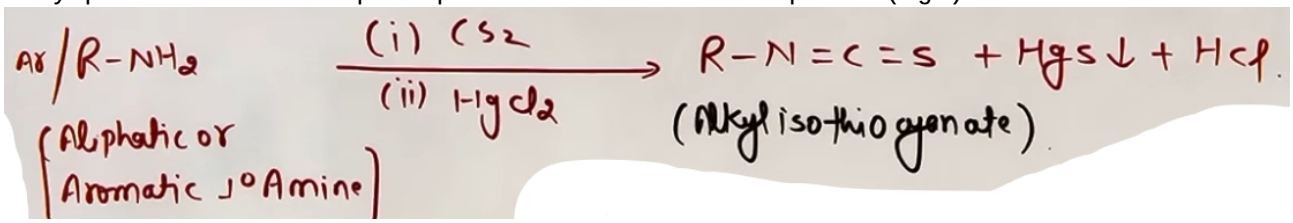
This is a diagnostic test used to identify primary (1°) amines.

REAGENTS AND PROCESS

- Reactants: 1° Amine (Aliphatic or Aromatic) + Carbon Disulphide (CS_2).
- Follow-up Reagent: Mercuric Chloride ($HgCl_2$).

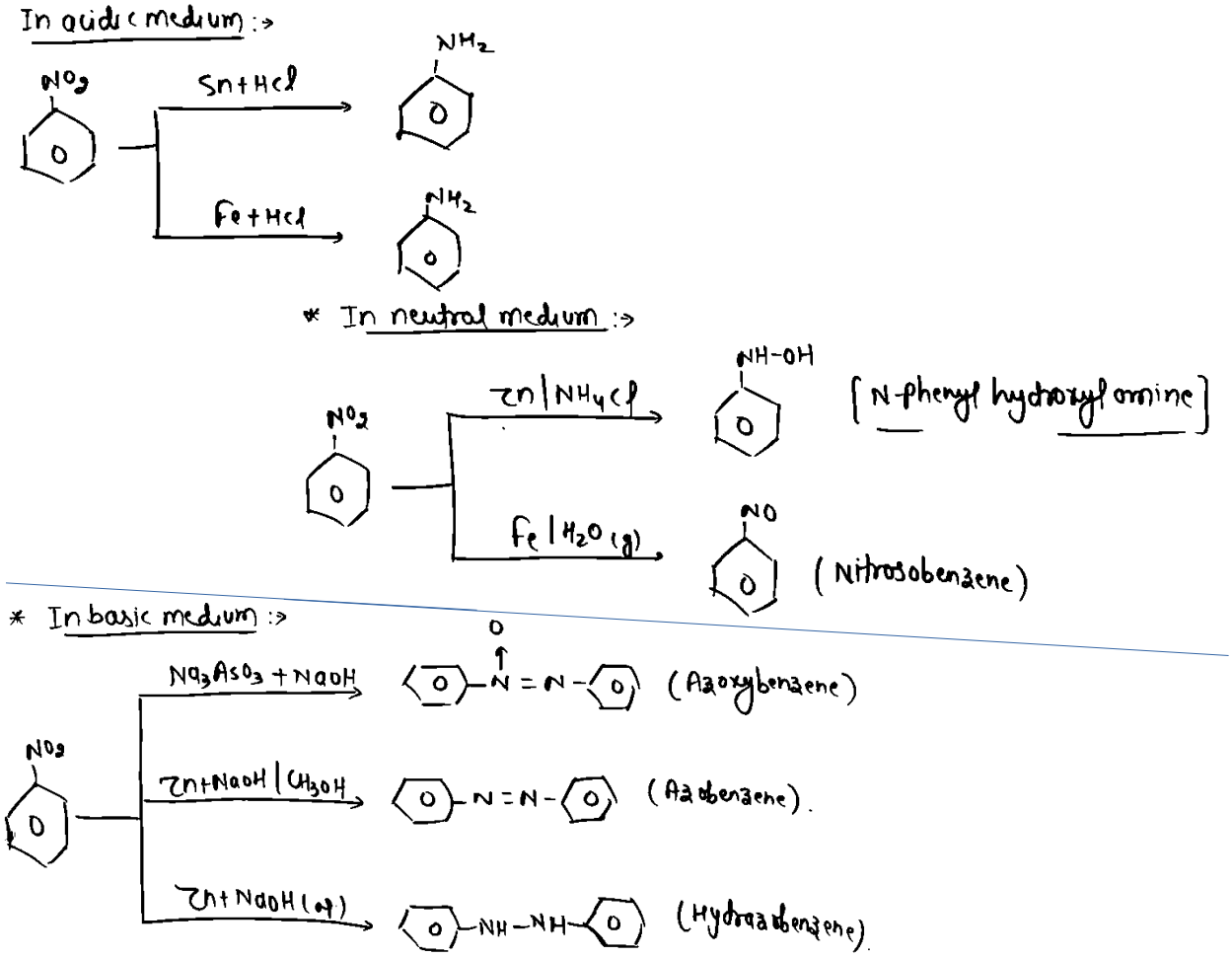


OBSERVATION
- Product: Alkyl Isothiocyanate (R-N=C=S).
- Smell: The product has a characteristic smell of Mustard Oil.
- By-product: A black precipitate of Mercuric Sulphide (HgS) is formed.

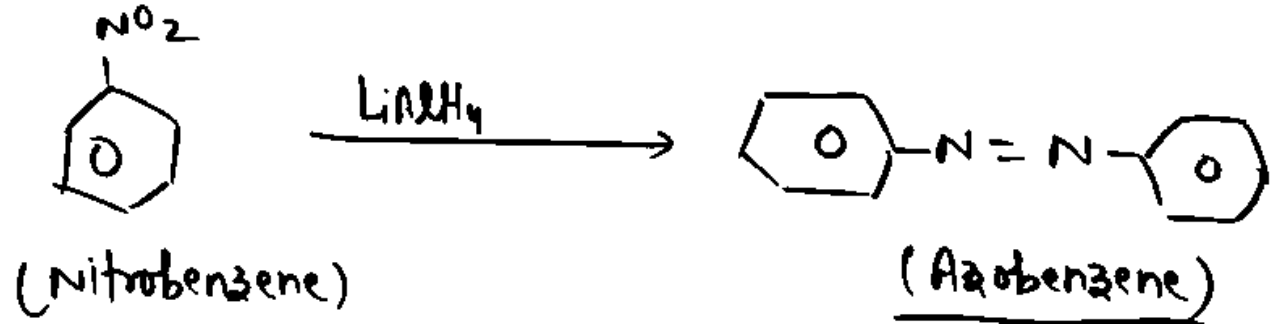


E. REDUCTION OF NITROBENZENE IN DIFFERENT MEDIA

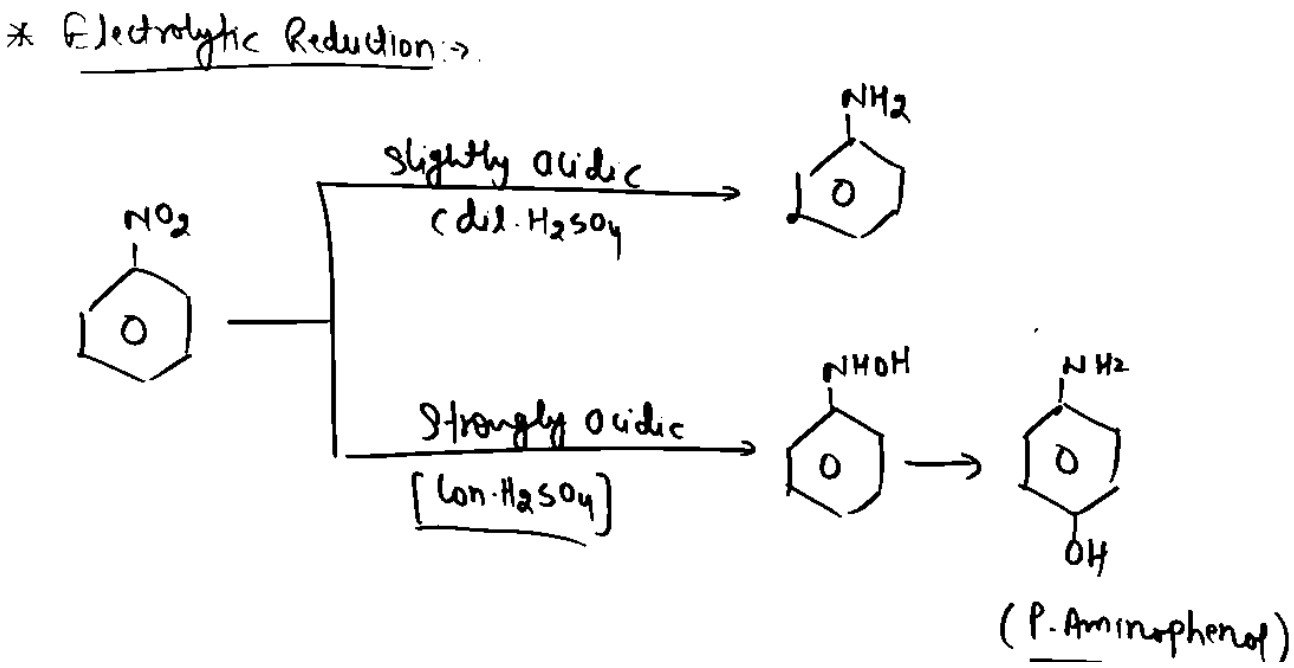
The reduction of Nitrobenzene ($C_6H_5NO_2$) is highly sensitive to the pH of the medium. The final product changes depending on whether the conditions are acidic, neutral, or basic.



F. REDUCTION WITH $LiAlH_4$



G. ELECTROLYTIC REDUCTION
- Slightly Acidic (Dilute H_2SO_4): Yields Aniline.
- Strongly Acidic (Conc. H_2SO_4): Initially forms Phenylhydroxylamine, which immediately undergoes rearrangement to form Para-aminophenol ($HO-C_6H_4-NH_2$).



PROBLEM-SOLVING FRAMEWORK

Question Type	Identification Clues	Approach / Strategy	Common Mistakes
Identify 1° Amine	Mustard oil smell / Black ppt	Hoffmann Mustard Oil Test (CS_2 + $HgCl_2$).	Thinking 2° or 3° give the test.
Alkene Product	Quaternary salt + Heat	Form the LEAST substituted alkene.	Picking the Saytzeff product.
Medium effect	Nitrobenzene + Zn/NH_4Cl	Neutral medium = Hydroxylamine product.	Writing Aniline as product.
Amide to Amine	Carbonyl removal	If C is lost: Hoffmann Bromamide. If C stays: $LiAlH_4$.	Swapping the reagents.

JEE ADVANCED INSIGHTS AND RANK-IMPROVING TIPS

HIDDEN OBSERVATIONS
- In Hoffmann Elimination, the leaving group ability and steric bulk are the deciding factors. If the leaving group is neutral (like in alkyl halides), Saytzeff is preferred. If it is charged (like $-NR_3^+$ or $-SR_2^+$), Hofmann is preferred.
- In the reduction of nitro compounds, Fe/HCl is preferred over Sn/HCl because $FeCl_2$ hydrolyzes to regenerate HCl, making it more economical.
- Many preparation methods (Gabriel, Hoffmann Bromamide, Curtius, Schmidt) are restricted to primary amines only.

TEACHER'S SUMMARY TIPS
- Name Reactions are King: Focus heavily on Hoffmann Bromamide, Gabriel Phthalimide, and Carbylamine tests.
- Basicity is Vital: Understand the aqueous phase basicity orders (213 and 231) thoroughly, as they are frequently tested in JEE Main.

HIGH-YIELD REVISION SHEET

MUST REMEMBER:

- Curtius/Schmidt Intermediate: Alkyl Isocyanate (R-N=C=O).
- Hoffmann Elimination: Least substituted alkene is Major.
- Mustard Oil Test: 1° Amine + CS_2 + $HgCl_2$ → R-NCS (Mustard smell) + HgS (Black).
- Nitrobenzene + Zn/NH_4Cl → C_6H_5NHOH (Neutral).
- Nitrobenzene + Conc. H_2SO_4 (Electrolytic) → p-aminophenol.

LAST-MINUTE REVISION POINTS:

- $NaNO_2$ + HCl at > 5°C → Phenol (from Aniline).
- Fe/HCl is better than Sn/HCl.
- 1° Aliphatic amines + HNO_2 → Alcohol + N_2 (Carbocation shift possible).
- Hinsberg Reagent: 1° (Soluble), 2° (Insoluble), 3° (No reaction).

$$$